

A COURSE IN INTELLECTUAL PROPERTY

THE RIGHT TO COPY

Copyright, Trademarks,
and Creative Work
in the Age of AI

*What can you protect when a machine
can say it differently—or build it again?*

BOOKS / SOFTWARE / TRUST

International foundations · National differences
Practical solutions and their limits

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Prologue — The competitor who copied nothing

Mara spends three years writing a book that teaches people how to reason about uncertain decisions. She invents a memorable sequence of examples: a flooded archive, an unreliable witness, a mistaken insurance claim, and a map that becomes less useful as it becomes more precise. Readers tell her that the examples finally made a difficult subject intelligible. A competing book appears. Its sentences are different. Its central journey is strangely familiar. The same misconceptions arrive in the same order. Even the archive has survived, although it is now a basement laboratory.

Adrian has a different problem. His small software company has spent years building a reliable application for organizing research evidence. A competitor releases an application with the same useful features. Its code is different, its name is different, and its developers say that an AI coding agent wrote it. Customers ask why they should pay Adrian more. He asks a lawyer whether years of engineering effort must count for something.

These are fictional situations, not disguised accounts of real disputes. They will accompany us through this book because they expose a question that slogans conceal: what exactly was taken? A sentence, a sequence of creative choices, a method, a reputation, a customer, an opportunity, or merely the advantage of having arrived first? Those are not interchangeable losses.

A copyright rule can protect the expression in Mara's book without giving her ownership of rational decision-making. A trademark can help customers distinguish Adrian's application from an impersonator without granting him the exclusive right to organize research evidence. That separation is foundational, not a loophole discovered by artificial intelligence. Copyright protects certain expressions; trademark law

protects certain source-identifying signs. Neither is a general promise that a useful product will remain difficult to compete with. [1, 3, 36]

Large language models make this uncomfortable distinction economically vivid. Suppose a machine can produce a convincing substitute for some of a book's practical value without reproducing its protected expression. Suppose an agent can independently implement an unpatented function from a lawful specification. In those cases, better enforcement of existing copyright cannot necessarily restore the original creator's market advantage. Conversely, replacing words, changing variable names, or using a model does not establish that protected expression has disappeared. The central task is to investigate the relationship between the works, not to admire the transformation tool.

This book argues for a disciplined middle position. Creators should have effective remedies against unlawful appropriation and deception. Users should retain the ability to learn, criticize, interoperate, and compete. Technical systems should make the relevant evidence easier to preserve and examine. But society should resist converting all useful meaning into private property merely because machines have become better at translating meaning into new forms.

The journey begins with the basic concepts. It then follows rights across borders, explains trademarks as a system of trustworthy identification, examines books and software separately, and develops practical responses. The final chapter asks what institutions should do when lawful competition itself becomes more disruptive. The objective is not to replace a lawyer. It is to make a conversation with one precise enough to be useful.

Reader's orientation

This is an educational work, not legal advice for a particular transaction or dispute. Its research cutoff is 11 September 2026. The jurisdictional survey is selective: the international framework, the United States, the European Union, Romania, Switzerland, the United Kingdom, and Japan receive particular attention. It does not claim to catalogue every national exception or every pending AI case. For action, verify the current authoritative text, applicable national implementation, contractual terms, procedural deadlines, and later judicial developments with qualified counsel.

Numbered references identify legislation, judicial decisions, official guidance, technical standards, and selected research. They lead to the source register at the end. A treaty is not a national statute; a policy report is not a judgment; a settlement is not a general precedent; and a technical specification is not a guarantee of legal compliance. Those distinctions matter throughout the course. Historical decisions are dated. Recent policy statements are described as policy, not silently promoted into law.

Mara and Adrian, their organizations, their works, and all uncredited teaching examples are invented for this book. Example names have not been cleared as trademarks. Illustrative figures are assumptions, not industry measurements. Proposed workflows and policy mechanisms are analytical recommendations, not existing legal safe harbours.

The manuscript was researched and drafted with AI assistance. No claim is made that every sentence is a human-authored contribution or that AI output automatically attracts copyright. Publication under a human or organizational name should follow substantive human editorial review and an accurate assessment of rights. This disclosure is especially important in a book about the difference between directing a tool, accepting its output, and creating protectable expression.

1 — What, exactly, can you own?

A copyright is not ownership of everything a reader learns.

1.1 — Why a copy is not a stolen chair

Imagine two different intrusions into Mara's office. Someone takes her only chair. She no longer has a chair. Someone photographs a page of her manuscript. She still has the page, but she may have lost control over an economically important use of it. Both acts can be wrongful; they are not wrongful for the same reason. The difference explains why intellectual property needs carefully delimited rules instead of a simple transplantation of rules about physical possession.

Copyright is a legally defined set of exclusive rights over qualifying works. International instruments recognize economic rights and, through the Berne framework, moral rights associated with authorship and integrity. The protected object is a work expressed in a legally relevant form, not every idea or useful result connected with it. Protection normally arises without registration in the Berne system, although national procedures can greatly affect enforcement. [1, 2]

A physical object and a work can therefore have different owners. A museum may own a painting without owning the right to reproduce it commercially. Mara's reader owns a printed copy without becoming the copyright owner of the text. Adrian's customer may receive permission to use software without acquiring its source code or the power to redistribute it. Thinking in layers is less intuitive than thinking in objects, but it prevents expensive mistakes.

There are several ways to justify these rights. An incentive argument asks whether creators and publishers will invest when competitors can copy cheaply. A personality argument emphasizes the connection between expression and the person who created it. A fairness argument

asks whether others should profit from someone's labour without consent. An institutional argument asks what arrangements produce a rich, accessible, trustworthy culture. These are different arguments. They can support different durations, exceptions, and remedies. None makes unlimited control over every influence a necessary conclusion.

Consider an imagined town with one hundred readers. Mara will write a specialized book only if she expects enough revenue to cover her time. Cheap verbatim copying could undermine that expectation. Yet the town also needs teachers who explain the subject differently, critics who challenge her conclusions, libraries that preserve books, and researchers who apply what they learn. Protecting Mara too little could reduce future books. Protecting her too much could make the first book the last authorized starting point for everyone else. The relevant goal is not maximum exclusivity but a defensible balance between creation and subsequent use.

That balance cannot be settled by calling one side selfish. A reader who cannot afford a book has a real interest in access. A writer who cannot afford to continue writing has a real interest in compensation. A developer who wants to build a compatible tool may be serving customers trapped by an incumbent. An incumbent who has invested in careful documentation may reasonably object to literal repackaging. An educational discussion must admit these interests before deciding which legal instrument can address them.

The history of the international system also matters. Berne, adopted in 1886 and subsequently revised, helped create protection across national boundaries. The later WIPO Copyright Treaty addressed the digital environment, including communication and technological measures. These instruments are not evidence that every new technology must be prohibited; they show that institutions have repeatedly adjusted the relationship between dissemination and control. [2, 4]

A useful classroom exercise is to ask what would happen without a particular restriction. Would the disputed act substitute for a licensed edition? Would it enable criticism? Would it reveal a fact? Would it let

a competing program read the user's own files? The question does not determine legality on its own. It reveals the interests that a legal rule is supposed to reconcile.

Adrian's frustration now becomes more precise. "We worked hard" is a morally relevant claim, but not a complete legal theory. He must identify an asset and a protected act: copied source code, confidential material improperly obtained, a patented invention, misleading branding, or a breached agreement. Years of work do not become one undifferentiated right called ownership of the product category. The discipline of specifying the right is the beginning of both effective protection and legitimate competition.

1.2 — Originality, expression, and the layers of a work

A book about probability contains facts, mathematical relationships, terminology, examples, diagrams, sentences, and a teaching sequence. Some of these may be protectable expression; some are not. A computer program likewise contains algorithms, constraints imposed by its environment, conventional structures, and expressive implementation choices. The fact that the finished object is protected does not make every element inside it exclusive.

The United States Supreme Court's decision in *Feist* explains originality as independent creation with a minimal degree of creativity, not mere effort or novelty. Facts themselves are not created by the person who discovers them. An original selection or arrangement can be protected without converting the underlying facts into property. The case rejected protection for the ordinary white-pages listings at issue. This is a United States decision, not a universal formula for every national originality test. [54]

Suppose Mara writes: "The witness remembered the red car because the photograph showed a red car." Another writer explains that later information can alter a witness's recollection. The second writer has used an idea or factual proposition. Now suppose the second writer reproduces Mara's invented village, the witness's peculiar mistake, the

staged sequence of revelations, and the final reversal, merely replacing red with blue. That looks different from independently explaining the general phenomenon. The issue has moved from information to the creative way in which the information was embodied.

The same analysis works at several levels. At the lowest level, strings of words may match. Higher up, paragraphs may reproduce a distinctive progression. Higher still, an elaborate fictional scenario or selection of unusual teaching examples may have been appropriated. At a sufficiently abstract level, however, what remains is often an unprotected theme, topic, technique, or method. Legal analysis must find the relevant level rather than simply choosing the one most favourable to the speaker.

This is why both popular shortcuts fail. “The words differ” ignores nonliteral expression and adaptation. “The meaning is the same” ignores the freedom to communicate ideas. Neither conclusion follows from a single similarity score. A legal comparison asks what was copied, whether it was protected, whether enough was taken under the applicable test, and whether permission or an exception applies.

Fixation adds another distinction. United States copyright generally requires an original work fixed in a tangible medium, and its statute excludes ideas, systems, procedures, processes, and methods from protection as such. Other national systems have their own requirements and may protect some unfixed expression. A practical creator should preserve a stable version rather than assume that an undocumented conversation will be easy to prove later. [8]

Derivative works make the importance of expression especially clear. A translation changes almost every word while remaining recognizably a version of the earlier work. An adaptation can move a narrative into another medium. The addition of creative material does not necessarily extinguish the rights in the material adapted. Conversely, a new work about the same topic is not automatically a derivative work merely because it learned something from its predecessor. The legal categories

concern a particular relationship, not the fact that culture has influences. [8]

A useful method is to mark a disputed passage in three colours on a private working copy: facts and methods; conventional or constrained elements; and potentially distinctive expressive choices. This is an analytical exercise, not a courtroom algorithm. The colours force the analyst to explain why an element belongs in one category rather than another. If everything receives the “protected” colour, the analysis probably has not begun. If nothing does, the analyst may be confusing a work’s factual subject with the author’s treatment of it.

For Adrian, the exercise could identify a standard file format, a required protocol sequence, a public algorithm, a peculiar decomposition into modules, and original documentation. For Mara, it could distinguish a commonplace definition from an elaborate explanatory scene. In each case the commercially valuable part may not coincide with the legally exclusive part. That mismatch is one of the central difficulties of creative enterprise, and AI intensifies it rather than inventing it.

1.3 — The bundle of rights and the chain of ownership

Mara’s publisher sends a contract that says it wants “all rights in all media.” The phrase sounds efficient. It is also a warning to slow down. A printed edition, an audiobook, a translation, a film adaptation, a searchable database, and a model-training licence are not the same commercial opportunity. Treating them as interchangeable can surrender options that neither party has priced or understood.

Ownership and authorship are separate questions. Under United States law, copyright initially belongs to the author, subject to statutory rules such as work made for hire; ownership of a physical copy is distinct from copyright ownership. Transfers of copyright ownership generally require a signed writing. Joint ownership, employment, commissioned work, and corporate ownership introduce special rules. A contractor is not automatically an employee, and labelling a contract “work for hire” does not remove the statutory conditions. [9]

A licence permits specified uses while leaving ownership where the agreement places it. An assignment transfers identified rights. An exclusive licence may prevent even the licensor from authorizing another party within its scope; a nonexclusive licence can coexist with other permissions. The operational questions are concrete: which work, which rights, which territory, which language, which duration, which media, which sublicensing powers, and which payment mechanism?

Imagine that Mara granted only English print rights five years ago. A later publisher wants to supply the entire manuscript to an AI service that produces personalized study guides. The earlier contract's wording, the relevant law, and the proposed operations must be examined. It is not enough to say that both uses involve a book. Nor should Mara assume that the word "digital" in an old contract definitely excludes or definitely includes every future computational use. An amendment that names the intended activity may be safer than litigation over a vague phrase.

Moral rights create another layer. Recognition of authorship and protection against certain harmful distortions are not simply payment claims. Their scope, duration, transferability, and waivability vary significantly. A broad economic assignment does not automatically resolve them. The Berne framework recognizes attribution and integrity interests, while domestic law supplies important details. [2]

For a practical rights ledger, the unit of analysis should be the component rather than the product. A book can contain commissioned illustrations, licensed photographs, a translator's contribution, quoted material, and a separately designed cover. A software package can contain employee code, contractor code, open-source dependencies, fonts, sample datasets, and documentation. "Our company owns the product" is not a substitute for tracing each relevant permission.

Consider Adrian's acquisition review. A contractor created a central module before the company existed. The founder later paid an invoice but never obtained a clear assignment. A contributor included a library with obligations the team did not record. A designer supplied a logo but

retained copyright in the artwork. None of these facts proves that the business cannot operate. They show why a buyer will ask for evidence rather than accept a confident ownership declaration.

Duration is equally easy to misunderstand. For many modern United States works, the general term is the author's life plus seventy years; special categories, including qualifying works made for hire, use different calculations. Older works can require a separate historical analysis. Copyright is not renewed indefinitely like a maintained trademark. A publication date alone often cannot tell a reader whether a particular edition is free to reuse. [10]

The right habit is to treat the chain of title as a living record. Keep signed agreements, their amendments, the covered versions, licences that arrived with dependencies, and evidence of the capacity in which contributors worked. Record uncertainty openly. A ledger with three unresolved entries is more useful than a folder labelled "all cleared" whose contents have never been examined. For AI-assisted work, include the human contribution and the permitted uses of the supplied inputs. Ownership begins to become manageable when it is a traceable relationship rather than an assertion.

1.4 — Permission, exceptions, and the public domain

A lecturer wants to criticize a paragraph. A reviewer wants to display a small image of a cover. A library wants to preserve a deteriorating work. A student wants an accessible format. These are different activities, and legal systems do not treat "using someone else's work" as one uniform transaction. Some uses require permission; some fall within exceptions; some involve no protected material at all.

United States fair use considers four statutory factors: purpose and character, nature of the work, amount and substantiality, and effect on the relevant potential market or value. It is a contextual analysis, not a numerical allowance or a rule that educational use always wins. The Copyright Office's index illustrates how outcomes depend on facts. Other jurisdictions often use more specifically defined exceptions or

fair-dealing categories rather than importing this open-ended United States doctrine. [57]

Suppose a reviewer quotes a short passage because its wording is the object of criticism. Compare a subscription service that gives readers the most memorable pages of each new book so they need not buy it. Both may claim to educate. Their purposes, amounts, and relationships to the original market differ. Now compare an independent explanation of the book's general thesis that takes no protected expression. That third activity may not need a fair-use defence for the thesis at all. Before analysing an exception, identify what exclusive right would otherwise be implicated.

The Supreme Court's Warhol decision illustrates the danger of treating new meaning as a universal defence. The dispute concerned a particular commercial licensing use of an image derived from a photograph. The Court's analysis of the first fair-use factor did not make every alteration transformative merely because it conveyed something different. The holding must be read in relation to that use, not as a general prohibition on artistic adaptation or a complete ruling about every use of the artwork. [17]

Attribution does not replace permission. A copied chapter with an impeccable bibliography can still be unauthorized. Equally, a legally permissible use can be academically dishonest when it hides intellectual dependence. Copyright infringement and plagiarism overlap but answer different questions. The first concerns legally protected rights and acts; the second concerns norms of credit, honesty, and scholarly contribution. This distinction is essential when a model paraphrases an argument without naming its source.

The public domain is not a low-quality corner of culture. It is the space in which exclusive copyright either does not arise or has ended, subject to jurisdictional and component-specific qualifications. The underlying text of an old work may be free while a recent translation, annotation, photograph, or recording has separate rights. A platform's access terms may raise additional questions without recreating expired

copyright in the underlying work. A publisher should distinguish the public-domain work from the particular file or edition it obtained.

Digital locks complicate this picture. United States law separately regulates circumvention and certain dealings in circumvention technology, with statutory exceptions and periodically defined exemptions. It also addresses certain misconduct involving copyright-management information. The practical lesson is not that every lock creates unlimited rights, but that a permitted downstream purpose does not automatically answer every question about how access was obtained. [13]

Mara can publish a sample chapter under a clearly defined licence while reserving other uses of the full book. A reader can distinguish quoting that sample, sharing it under its licence, and uploading the entire commercial edition to an external service. Adrian can permit interoperability testing without permitting redistribution of his source code. Specific permissions often reduce friction better than absolute slogans.

The classroom test is simple to state and difficult to answer well: what is the legal basis for this use? “It was online,” “I paid for a copy,” “I named the author,” and “a machine did it” are not complete answers. A sound answer identifies an absence of protected subject matter, a licence, an exception, expiration of rights, or another relevant legal basis, and then states the factual conditions on which it depends.

1.5 — The other rights that people confuse

When someone says “protect my idea,” the necessary response is a question, not a form. Do they want to prevent code copying, preserve a confidential process, stop an impersonator, control a patented invention, or negotiate the terms of a collaboration? Choosing the wrong instrument can produce both wasted expenditure and exaggerated expectations.

Trademark law concerns signs that distinguish the source of goods or services. Its central orientation differs from copyright’s protection of expression. A name can be a powerful trademark even when the word itself is not a copyrighted work. A complex logo may involve both

trademark protection and copyright in the artwork. The two rights can belong to different parties and have different territorial and contractual histories. [36]

Patents, where available and valid, can protect claimed inventions against certain uses even when another party arrived independently. Copyright usually does not function that way. A reader who independently implements a general method described in a book is not thereby copying the book's expression. *Baker v. Selden* drew an enduring United States distinction between copyright in an explanation and exclusive rights over the system explained. It is a historical decision, not a complete account of modern patent eligibility. [55, 58]

Trade-secret protection has another logic: information must satisfy requirements involving secrecy, commercial value linked to secrecy, and protective measures. The EU Trade Secrets Directive distinguishes unlawful acquisition from routes such as independent discovery and, under specified conditions, observation or testing of lawfully possessed or publicly available products. It does not create copyright in every useful confidential fact or remove all contractual questions. [27]

For Adrian, secrecy may suit a server-side optimization that competitors cannot readily observe. It is a poor strategy for a feature visible to every user. Publishing a detailed explanation to attract customers and simultaneously assuming nobody can learn the method from it is a contradictory plan. The decision to disclose should therefore be made consciously, with consideration of what can remain protectable after disclosure.

Contracts allocate obligations between parties. They can regulate access, confidentiality, permitted processing, audit, payment, and risk. But a contract is not a universal statute binding everyone who later encounters an idea. Enforceability depends on formation, applicable law, mandatory rules, and the particular restriction. The practical recommendation is to describe the intended transaction precisely and avoid assuming that broad boilerplate makes difficult legal questions disappear.

Databases illustrate why a single “copyright cleared” label is insufficient. A collection may contain individually protected works, protectable selection or arrangement, access restrictions, personal data, and—in the EU—a distinct database-right regime. The specific legal interest at stake must be identified. A person may be free to state a fact while still facing questions about systematic extraction from a protected database or misuse of confidential access. [3, 59]

Now place Mara’s book beside Adrian’s application. The text, artwork, code, name, user data, confidential methods, and contractual relationships form separate layers. Some can be licensed openly; some can be kept private; some cannot be owned in the desired way. A realistic strategy protects the defensible layers and builds value in the others. It does not inflate a weak claim until it sounds comprehensive.

At the end of this first chapter, the reader should be able to reformulate an emotional complaint into an analysable proposition. Not “they stole our product,” but “they reproduced this protected module without the required permission.” Not “they stole my knowledge,” but “their book preserves these distinctive invented examples and their expressive arrangement.” Not “they used our word,” but “their presentation is likely to mislead relevant customers about origin or affiliation.” Precision does not make every claim succeed. It makes success, compromise, and legitimate competition distinguishable.

2 — One internet, many laws

A file can cross a border without carrying the law of its birthplace.

2.1 — The international architecture is a network, not a world code

Mara publishes in English from Romania. A Swiss distributor sells her book. A United States service processes it on infrastructure located elsewhere and answers a customer in Germany. Which copyright law applies? There is no useful answer consisting only of the place where the website was registered. The acts, locations, claims, defendants, contracts, and forum must first be separated.

Berne provides a framework of national treatment, automatic protection, and minimum standards. National treatment broadly means that eligible foreign authors receive the protection a country gives its own nationals, subject to the Convention's rules. It does not mean that the author carries one home-country copyright law everywhere. TRIPS adds trade-related minimum obligations and enforcement disciplines, incorporates much of Berne, and expressly distinguishes expression from ideas and methods. TRIPS does not incorporate Berne's Article 6bis moral-rights obligations, which should not be mistaken for abolishing moral rights in national law. [2, 3]

The WIPO Copyright Treaty supplements the framework for the digital environment. Its subject matter includes computer programs and original compilations, and it addresses distribution, communication, technological protection, and rights-management information. These international obligations are implemented and interpreted through legal systems. A treaty can make national rules more alike without erasing differences in exceptions, ownership, remedies, or procedure. [4]

For trademarks, the Paris Convention establishes important principles including national treatment, priority, and protection against unfair competition. Its priority mechanism can preserve an earlier filing date for subsequent filings within the applicable period—six months for trademarks—when the relevant conditions are met. It is not an automatic worldwide trademark. [5]

The Madrid System is a procedural route for seeking trademark protection in designated participating jurisdictions through an international application and registration framework. Designated offices retain substantive examination powers. Refusal in one jurisdiction does not mean the mark has necessarily failed everywhere, and acceptance elsewhere does not force a country to ignore its own grounds of refusal. The system simplifies administration; it does not turn the world into one trademark territory. [6]

Think of the international system as a railway network. Agreements help trains move between national systems, but the stations still have rules, and a ticket to one destination is not a ticket to every destination. This analogy also identifies a common commercial mistake: paying for an international filing without identifying the markets in which protection is actually needed.

For Mara's book, the analysis could begin with the disputed reproduction and the market where relief is sought. For Adrian's trademark, it could begin with the territory in which customers encounter the sign and the rights registered or acquired there. For their contracts, a governing-law clause and a forum clause may matter without deciding every noncontractual claim. Those are questions for jurisdiction-specific advice, not details that disappear because the service is delivered online.

There is also an enforcement budget. Winning a legal argument in principle does not guarantee affordable evidence collection, a solvent defendant, or practical removal of distributed copies. An international strategy should therefore identify not only rights but leverage: contractual counterparties, app stores, payment relationships, commercial distributors, and identifiable hosting operators. Those channels must be

used truthfully and proportionately. Sending a platform a misleading infringement allegation is not a legitimate substitute for having a right.

An instructive exercise is to draw three maps for a project: where relevant content is obtained and processed, where customers are targeted, and where enforceable counterparties or assets exist. Differences between the maps reveal legal and operational questions. A project does not become lawful merely by placing one server in a favourable jurisdiction. Nor does every globally accessible website necessarily trigger identical consequences in every country. The point is to replace a fictional universal jurisdiction with a documented analysis of real acts.

2.2 — The United States: flexible exceptions, demanding distinctions

The United States is important in AI disputes because many influential services, contractual relationships, and litigated acts connect to it. That importance does not make its rules global. A European publisher who says “fair use permits this” must still explain why United States law governs the relevant act and whether the asserted use actually satisfies the doctrine.

Several practical distinctions deserve attention. Registration is not what initially creates ordinary United States copyright, but registration or refusal is generally required before bringing an infringement action concerning a United States work, subject to statutory qualifications. Timely registration can affect eligibility for statutory damages and attorney’s fees. These procedural consequences are different from the underlying existence of copyright. A foreign work can raise different registration prerequisites while still presenting remedial questions. [11]

Mara should therefore not assume that a copyright notice is an enforcement plan. She needs to understand whether and when registration is useful for the works and markets at issue. A notice can communicate a claim; it does not automatically establish ownership, human authorship, or compliance with registration requirements. Filing an inaccurate

description of AI-generated content could create difficulties rather than solve them.

Fair use is particularly relevant to analysis, search, criticism, and computational processing, but “transformative” is not a synonym for technologically sophisticated. A tool that uses a text for a different function may have arguments unavailable to a tool that simply delivers a competing readable substitute. Still, purpose alone is not the complete statutory inquiry. Amount, the nature of the source, and legally cognizable market effects also matter. The existence of a hypothetical licence does not mechanically decide the analysis, and neither does an asserted public benefit. [57]

A further distinction separates litigation about training from litigation about output. The United States Copyright Office’s May 2025 prepublication report on generative-AI training discusses the issue as context-dependent rather than universally fair or universally infringing. The Office’s AI page still identifies that document as a prepublication version in the material consulted for this book. An agency report is informative, but it is not a Supreme Court judgment or an enacted general training exception. [15, 16]

For a business, uncertainty should produce a decision record. Suppose Adrian can choose between a model backed by well-documented training permissions and a cheaper model with opaque provenance. He should compare actual evidence, contractual protection, use restrictions, indemnity conditions, and the consequences of a dispute. He should not treat a vendor’s marketing claim that a model is “safe” as a substitute for examining the transaction. Conversely, incomplete public disclosure is not itself proof that a model infringes.

United States remedies also require careful reading. Injunctions, actual damages and attributable profits, and statutory damages operate under specific requirements. Statutory ranges are not automatic invoices, and counting outputs is not necessarily the same as counting works for damages. Safe-harbour provisions for qualifying service providers have conditions and do not declare every hosted item lawful. An

enforcement strategy that begins with the largest conceivable number can mislead both the rights holder and the alleged infringer. [12]

The practical lesson is neither “litigation is useless” nor “register and you are protected from AI.” It is to align procedural preparation with the actual asset. Preserve drafts, versions, publication facts, relevant contracts, and evidence of the disputed use. Distinguish direct copying from a general competitive threat. Document the human contribution accurately. Keep a route for resolving small issues without escalating them into expensive symbolic battles.

For the reader, the United States offers a methodological warning: flexible legal standards require more factual discipline, not less. A four-factor framework does not mean that any preferred outcome can be dressed in four paragraphs. It means that the advocate must explain why the particular use, of the particular material, in the particular market, fits the doctrine.

2.3 — The European Union: harmonized layers and two mining exceptions

A European project does not face one undifferentiated “AI copyright law.” It faces copyright rules, software-specific rules, national implementation, trademark law, and AI-provider obligations that answer different questions. Confusing those layers produces claims such as “the AI Act authorizes training” or “a no-AI notice prohibits every use.” Neither is a reliable statement of the framework.

The Software Directive protects the expression of computer programs rather than their underlying ideas and principles. Other directives address reproduction and communication, databases, duration, and enforcement. The EU’s general harmonized term for an author’s copyright is life plus seventy years, with special rules for particular categories. Member States retain important national rules within and alongside harmonized fields. Romania, France, and Germany are therefore not interchangeable for every ownership, moral-rights, contract, or procedural question. [20, 26]

The DSM Directive’s Articles 3 and 4 create distinct text-and-data-mining routes. Article 3 concerns eligible research organizations and cultural-heritage institutions conducting scientific research with lawful access. Article 4 is broader in beneficiaries and purposes but is subject to an appropriate express reservation of rights; online reservations are contemplated in machine-readable form. Both have conditions, and their scope must be examined for the particular acts. The Article 3 route is not displaced merely by an Article 4 reservation. [21]

The difference can be made concrete. An eligible university team with lawful access studies patterns across publications for a scientific project. A commercial service mines a catalogue to develop a product. They cannot simply exchange legal labels because both operations involve algorithms. Nor can an organization claim the research route merely by naming its product department “the research laboratory.” Eligibility, purpose, access, security, and the actual operation require analysis.

A second misconception treats mining as the entire product lifecycle. Suppose access and copying for a qualifying analysis are permitted. That does not automatically authorize publication of a copyrighted chapter in the final user interface. Conversely, a carefully limited output does not settle whether the corpus was lawfully acquired. The permission or exception must correspond to the act, not to the project’s most attractive description.

The AI Act adds a provider-compliance layer. Article 53 requires general-purpose-model providers within its scope to have a policy for compliance with EU copyright law, including identifying and respecting Article 4(3) reservations, and to publish a sufficiently detailed summary of training content using the prescribed template. Certain open-source-model exemptions concern specified documentation duties, not a blanket exemption from the copyright-policy and summary duties. These are not private licences from every rights holder and do not themselves resolve infringement claims. [22]

The General-Purpose AI Code of Practice, published in July 2025, is a voluntary means of demonstrating compliance with relevant obligations, not a replacement for copyright law. The Commission’s materials consulted for this book distinguish that voluntary compliance route from the binding legal obligations. They also identify the phased application of the AI framework. For procurement, the important question is the provider’s applicable obligations and transitional status, not whether a code logo appears on a website. [23, 24]

Mara’s publisher can express reservations appropriately and preserve evidence of when and where they applied. That is an operational step, not a promise that every historical training event becomes reversible. Adrian can ask his suppliers how they implement relevant reservations and document downstream information. Neither should claim that a crawler signal alone supplies complete global legal protection.

For a European educational platform, a more useful design is a permissions matrix tied to operations: acquisition, storage, mining, retrieval, quotation, transformation, and distribution. A legal basis for one column should not automatically fill all the others. The platform can then explain a refusal precisely: “We may index this material under these conditions, but we have not established permission to return long passages.” That is better than either indiscriminate blocking or a universal “AI use permitted” flag.

2.4 — Romania and Switzerland: nearby markets, distinct rules

Adrian operates a Romanian development company and works with a Swiss business partner. Mara’s publishing project has readers in both countries. Geographic proximity and commercial familiarity can make differences look smaller than they are. Switzerland is not an EU Member State; EU directives do not become Swiss national copyright law merely because a project is European.

Romania’s principal copyright statute is Law No. 8/1996, with amendments including Law No. 69/2022. The consolidated text

available through WIPO Lex identifies the human creator as author, protects original literary, artistic, and scientific works, and distinguishes moral from patrimonial rights. Its software provisions include a default allocation of economic rights to the employer for programs created by employees in their duties or under employer instructions, absent contrary agreement. The statute also contains the research and broader mining provisions introduced in the European reform context. The text and its current amendments must be checked for a transaction. [29]

This default employment rule is highly relevant to Adrian, but it does not answer every collaboration question. Was the contributor actually an employee? Was the work created within the covered duties? Was there a contrary clause? Did the module incorporate pre-existing code? Was a separate contractor involved? A default rule can resolve one link in the chain while leaving several others open. Treating it as a universal declaration that the company owns all work near its employees is unsafe.

Mara should similarly separate recognition as author from the scope of a publishing agreement. A payment for an edition does not logically answer who can authorize a translation, produce a training corpus, or approve an adaptation. The relevant Romanian contractual provisions and the agreement's wording determine the result. Her operational response is to create a rights schedule in ordinary language before negotiating dense clauses.

Romanian trademark protection is governed principally by Law No. 84/1998, as amended. National rights coexist with the EU trademark system and relevant international registrations. A company name, a domain, a national mark, and an EU mark are different assets. A Romanian registration is not automatically a Swiss registration, and merely establishing a company does not establish that its chosen sign is safe for every proposed product. [30]

For an expanding small business, the sensible first question is not "How do we register everywhere?" It is "Where would a name conflict seriously disrupt our next stages of growth?" A business selling only

locally has different priorities from a software company already pitching enterprise customers in several countries. Clearance should precede expensive design and marketing commitments, not follow them as a ceremonial administrative step.

Switzerland offers a particularly memorable copyright distinction. The Swiss Federal Institute of Intellectual Property states the ordinary term as up to seventy years after the author's death, but fifty years for computer programs. It also identifies separate fifty-year protection from creation for photographs without individual character. Protection is automatic, with no copyright register. These differences show why importing one generic "life plus seventy" sentence into a global policy can be wrong. [31]

The point is not that Adrian should predict whether fifty rather than seventy years changes next year's revenue. It is that statutory categories matter. Once a team discovers a visible difference in duration, it should stop assuming that less visible rules on exceptions, employment, collective management, or enforcement are identical. A Swiss project requires a Swiss analysis, even when its staff have become fluent in EU terminology.

The Swiss trademark overview describes registration-based protection, renewable ten-year terms, and the requirement of genuine use, with non-use becoming relevant after the applicable five-year period. Search and monitoring remain necessary; registration is not an all-purpose guarantee against earlier rights. The national system must be considered separately from an EU filing. [32, 33]

A useful joint operating model would place Romanian employment and contributor documentation in one legal workstream, Swiss commercialization and local branding in another, and cross-border licences in a third. The purpose is not to multiply paperwork. It is to avoid using one country's answer for another country's question. A common internal evidence ledger can support all three while counsel applies the relevant law to each.

In the classroom, ask who owns a module created by a Romanian employee, contributed to a Swiss venture, and released with a third-party dependency. The correct initial answer is not a country name. It is a request for the employment facts, transfer documents, dependency licence, release conditions, and intended jurisdictions. The exercise teaches that national particulars are not trivia. They determine which missing fact can change the outcome.

2.5 — The United Kingdom, Japan, and the danger of exporting slogans

Two further examples are enough to demonstrate why “AI copyright” cannot be summarized as a global yes-or-no question. The United Kingdom has a distinctive statutory tradition concerning computer-generated works. Japan has a distinctive analytical exception associated with non-enjoyment uses. Neither can be reduced to an internet slogan without losing the qualifications that make the rule meaningful.

The United Kingdom government’s report of 18 March 2026 states that a broad mining exception with an opt-out is no longer its preferred approach and that further work is needed before reform. The report also discusses the existing computer-generated-works regime and possible change. These are policy positions, not proof that Parliament has already enacted the discussed repeal or a new general training permission. The current framework includes a narrower noncommercial-research mining exception, subject to conditions. [34]

This is a lesson in reading legal news. A consultation asks what should happen. A government response indicates policy direction. A bill proposes legal text. An enacted provision may still have a future commencement date. A judicial interpretation may narrow or clarify its application. Treating these documents as identical can make an internal compliance policy wrong before anyone has acted under it.

For Mara, the United Kingdom’s discussion of computer-generated works also challenges a simplistic international claim that every jurisdiction treats unassisted AI output exactly as the United States

Copyright Office does. Differences in statutory wording do exist. That does not establish that every prompted LLM book receives enforceable protection in Britain, or that the same result follows elsewhere. Identifying the human arrangements and the relevant originality requirements remains a serious interpretive task.

Japan's Agency for Cultural Affairs published a 2024 overview explaining Article 30-4 and AI. It describes non-enjoyment uses, including certain analytical activities, while emphasizing limits: mixed enjoyment purposes and uses that unreasonably prejudice a rights holder's interests can fall outside the provision. The document is expressly a nonbinding interpretation, not a definitive assessment of every AI technology. It also separates development and training from later generation and use. [35]

The conceptual interest is considerable. Reading a novel to enjoy it and analysing many novels to study linguistic patterns are different purposes. But what about training a system specifically to deliver close substitutes for one author's protected works? What about a database sold for analytical use? These questions test the boundaries of an exception rather than disappear beneath the phrase "Japan allows AI training." The actual source material and purpose must be examined.

The book does not attempt equivalent summaries for every other major economy. China, India, Canada, Australia, Singapore, and other jurisdictions have their own statutes, institutions, and developments. A short country list can create a misleading sense of clearance when it silently omits the one rule relevant to a particular deployment. The better transferable skill is to ask the same well-structured questions in each jurisdiction.

Start with the protected subject matter and the alleged act. Identify who performs it and where. Determine whether the source was lawfully accessed. Read the actual exception or licence. Separate copying for analysis from public output. Check ownership and human contribution. Then examine remedies and current procedure. These questions do not

guarantee one answer; they prevent the wrong answer from being imported by habit.

The most important international insight is therefore negative but productive: there is no reliable universal label saying “AI-cleared.” A project can instead maintain a set of scoped conclusions, each attached to evidence, a jurisdiction, an activity, a date, and an accountable reviewer. That is less elegant than a single green icon. It is also much closer to how the law actually works.

3 — Trademarks when everything can be imitated

A name can tell you who stands behind a promise. It cannot make the promise true.

3.1 — A trademark is an address for trust

Adrian's application could be reproduced in at least two importantly different ways. A competitor might build a different program that performs the same tasks and sell it under a clearly different name. Another might distribute an inferior program under a name, icon, and presentation that suggest Adrian's company made it. The first situation calls for an investigation of code, patents, contracts, and the permissible scope of competition. The second adds a source-identification problem. Customers may choose the wrong product because they misunderstand who is responsible for it.

A trademark is a sign capable of distinguishing one source's goods or services from those of others. Words and logos are familiar examples, but the legal categories can extend beyond them. The sign is not owned for every imaginable purpose: protection depends on the relevant law, rights, goods or services, and use. A company name, a domain registration, and a trademark registration are different things. Securing one does not establish entitlement to all three. [25, 36]

This difference from copyright is easiest to see with a plain word. An invented product name may have little or no copyright significance as text, yet function powerfully as a mark. An intricate illustration may attract copyright protection but fail to function as a trademark because customers see decoration rather than a source identifier. A logo can occupy both categories. Its designer's copyright and the business's trademark interest then need to be handled separately, including through a proper agreement.

Distinctiveness explains why some names make better candidates than others. The USPTO distinguishes stronger coined, arbitrary, and suggestive marks from descriptive expressions and generic terms. A descriptive term may acquire source-identifying significance; a generic term is the name of the relevant category rather than one undertaking's distinctive identifier. These categories depend on the goods and customer perception, not on the word in isolation. A familiar word can be arbitrary for an unrelated product. [37]

Suppose Adrian calls his tool "Research Evidence Organizer." The name efficiently describes its job. That marketing advantage comes with a legal disadvantage: other businesses also need ordinary words to describe the same job. A distinctive name followed by an explanatory subtitle may serve both purposes better. This is a strategic recommendation, not a clearance opinion about any particular candidate. The attractive invented word may already belong to someone in the relevant market.

Mara encounters the same issue when she creates a publishing imprint. A memorable imprint can identify the editorial institution responsible for a family of books. That identifier is conceptually different from the arguments in a book or the title of one publication. The distinction is useful because the imprint can persist while the content changes. It becomes an address to which readers direct trust, criticism, and expectations.

Trust must nevertheless be earned outside the register. A legally protected mark does not certify that a book is accurate, a program secure, or a model responsibly trained. It enables identification, after which ordinary evidence about quality still matters. Confusing registration with endorsement would repeat the very error that source-identifying law is meant to reduce.

For the AI economy, this suggests a limited but important opportunity. Where a substitute is easy to generate, knowing who reviewed it and who will correct it may become more valuable. That is an analytical expectation, not a guarantee of commercial success. Some

customers will prefer the cheapest adequate substitute. A trustworthy source must therefore offer something identifiable and useful, rather than merely insist that its name deserves loyalty.

3.2 — Clearing, registering, and maintaining a mark

The least expensive moment to discover a naming conflict is before a team has printed the packaging, published the documentation, and persuaded customers to remember the name. Trademark planning should therefore begin as a search problem, not as a graphic-design problem. The relevant search is broader than asking whether an identical string appears in a register. Similar pronunciation, appearance, meaning, and related markets can matter. The USPTO's guidance treats similarity between marks and relatedness of goods or services as central to confusion analysis. [38]

A practical clearance process begins with the actual business plan. What will be sold? To whom? Where? Through which channels? A name acceptable for an unrelated industrial product may conflict with an existing educational-software brand. A company that begins with an ebook and later offers courses, downloadable applications, and hosted services should discuss that realistic expansion with a trademark professional. Filing the vaguest possible description is not a substitute for doing this work.

The Nice Classification organizes goods and services for registration. It is an administrative framework, not a partition of commercial reality into unrelated legal universes. The EU trademark regulation expressly prevents treating goods or services as similar or dissimilar merely because they occupy the same or different Nice classes. A book, an educational application, and an online course may be commercially related even when their classification differs. [7, 25]

Territory is equally important. A Romanian national mark and an EU trademark have different geographic reach. Switzerland requires its own analysis. The Madrid System can simplify seeking protection in designated members, but it does not turn the application into a single

globally enforceable right exempt from local examination. Decisions about filing should follow expected markets, exposure, and budget rather than the ambition to “own the name worldwide.” [6, 30, 33]

An imagined small publisher might prioritize its home market and principal export markets, reserve appropriate domains, and monitor a narrower set of likely conflicts. A business preparing a major international launch might justify a more extensive search and filing programme. Neither plan is automatically superior. The right level of investment depends on the cost of renaming, likelihood of confusion, commercial importance of the identifier, and resources available for maintaining the portfolio.

Registration is also not a one-time purchase of immortality. Renewal and genuine-use requirements matter. EU registration runs in renewable ten-year periods, while a continuous five-year period of non-use can make a mark vulnerable, subject to the applicable rules and proper reasons. The Swiss IPI likewise describes renewable ten-year protection and the importance of use. Exact evidence, deadlines, and remedies must be checked in the relevant system. [25, 32, 33]

Adrian should preserve examples of real use: product pages, invoices, release materials, and dated packaging or screenshots showing the mark in relation to the relevant goods or services. The records should reflect reality, not manufacture an appearance of activity. He should also maintain ownership and licence documentation. A portfolio becomes less useful if the registered owner is a dissolved company while the actual business assumes somebody else handled the transfer.

The practical lesson is administrative rather than glamorous. A good mark is a maintained relationship among a sign, a responsible undertaking, a market, and evidence. The filing certificate is one important document within that relationship. It is not the relationship itself.

3.3 — Confusion, comparison, and the right to name a competitor

Imagine three advertisements. One says, “Our independent application imports files exported by Adrian’s application.” A second says, “The official successor to Adrian’s application,” although there is no such relationship. A third compares the two tools using a confusingly similar logo and an ambiguous download button. The presence of a competitor’s name does not make these advertisements equivalent. Their factual accuracy, presentation, and likely interpretation differ.

Trademark law must leave room for truthful references. A repair service may need to identify the products it repairs. A reviewer must be able to identify the book being reviewed. A compatibility statement may need to name another program. EU law includes limitations concerning necessary reference to the proprietor’s goods or services, subject to honest practices. US doctrines and their application are not identical. A responsible course should not invent a universal “nominative fair use” checkbox that resolves every jurisdiction. [25]

For practical drafting, the relevant questions are clear even before a lawyer assesses the final text. Is the claim true? Is the reference necessary to explain the relationship? Does the overall presentation imply authorization, sponsorship, or a commercial connection that does not exist? Is the competitor’s branding more prominent than needed? Could a reader identify the independent supplier without searching for a disclaimer? These questions improve design; they do not replace the legal test.

Disclaimers deserve particular skepticism. A small sentence at the bottom of a page may not repair a misleading headline or download flow. Conversely, an accurate and proportionate reference should not be assumed unlawful merely because it names another business. The evaluation must concern the whole use, not an isolated token. This matters for AI-generated interfaces, where a model can create a polished but misleading combination of familiar names, logos, and confident assertions.

Marks with reputation can receive protection extending beyond ordinary confusion in circumstances defined by the applicable law. EU rules address certain uses taking unfair advantage of, or harming, the distinctive character or reputation of a protected mark. That is not a rule that every famous business owns every cultural reference to its name. The requirements and limitations remain important. [25]

Mara’s book reviewer illustrates the public-interest side. A critical essay should be able to say which book it criticizes. A teaching page should be able to compare competing interpretations. Treating every unfavorable reference as infringement would make commercial identifiers a tool for silencing scrutiny. The creator’s legitimate interest in avoiding deception must coexist with the public’s legitimate interest in discussing recognizable objects.

An AI assistant introduces an additional design challenge. It may describe a third-party service as “official” without evidence. It may place a generated product beside the original brand in a way that suggests endorsement. Those failures are not solved by measuring textual originality. A platform should preserve the source of affiliation claims, require verification before applying official labels, and distinguish quotations from its own assertions. These are proposed safeguards against avoidable confusion.

The operational insight is that brand safety cannot be reduced to a blacklist of words. A system that blocks all competitor names prevents lawful comparison. A system that permits every mention can enable impersonation. Better design represents relationships: made by, compatible with, reviewed by, licensed by, inspired by, and unaffiliated with. Each relationship should mean something testable. A trustworthy interface should not use them interchangeably.

3.4 — Book titles, author identities, and synthetic endorsement

Mara initially believes that registering her book title will protect the entire book from competing versions. The belief combines two mistakes.

A mark does not give copyright-like control over content. Moreover, in US registration practice, the title of a single creative work generally does not function as a registrable trademark. A genuine series title can be different. The USPTO explains that editions or formats of the same work do not simply create a series, and that some categories, including computer software, require different treatment. [39]

This rule should not be generalized into “book titles are never protected anywhere.” National law and other causes of action can matter. Nor should a publisher manufacture a nominal series and assume that the underlying source-identification requirement has disappeared. The useful strategic question is whether customers recognize the sign as identifying an ongoing source, rather than merely identifying one expressive work.

Suppose a seller releases a book called “Mara’s Official Companion,” places Mara’s photograph on the cover, and says she approved the advice. The content might contain no copied paragraph. The problem would not thereby vanish. False endorsement, misleading presentation, trademark issues, personality interests, and other national rules may need separate examination. Copyright clearance cannot establish permission to appropriate someone’s commercial identity. A photograph may also carry its own rights.

Now change the facts. An independent scholar publishes an unmistakably independent critical guide to Mara’s work. Mara dislikes it, but the title accurately identifies its subject and the presentation does not suggest endorsement. The ethical and legal analysis must change. The ability to discuss another person’s work is essential to scholarship and criticism. A creator’s dislike is not, by itself, evidence of a protectable monopoly over that discussion.

AI systems make the difference between these examples particularly easy to blur. A model may adopt a familiar author’s manner, generate an invented endorsement, or produce a fake bibliography entry. A publisher may then mistake fluency for authorization. A sensible editorial rule is to treat every claim about a real person’s participation as a

factual claim requiring evidence. “The model said so” establishes only that a model produced text.

The same reasoning applies to a software product that names a respected engineer as an adviser or claims compatibility certification from a platform. These claims are about relationships in the world. They need documentary support independent of the model that generated the marketing copy. A signed statement can help identify the person making an assertion, but the publisher must still check its meaning, scope, and continuing validity.

Mara can strengthen her publishing practice without trying to monopolize critical discussion. She can maintain an authoritative catalogue of her own editions and approved collaborations, make corrections publicly traceable, and give retailers a clear route for reporting impersonation. Readers can then distinguish the genuine source from an unaffiliated interpretation. This is not perfect protection: false listings can still appear, and readers can still ignore the catalogue. It is a practical reduction in avoidable uncertainty.

The deeper issue is not whether an AI can sound like an author. It is whether the surrounding institution represents authorship, approval, and responsibility honestly. A persuasive imitation may be lawful in one respect and deceptive in another. Treating “AI-generated” as a complete legal category hides that distinction precisely when readers most need it.

3.5 — Why trademark cannot repair every copyright disappointment

The temptation is understandable. If copyright does not prevent a competitor from using public-domain material or unprotected ideas, perhaps a claim about “origin” can restore control. In *Dastar v. Twentieth Century Fox*, the US Supreme Court rejected a use of the Lanham Act that would effectively convert a dispute about authorship of communicative content into a new copyright-like entitlement. Its reasoning distinguishes the origin of tangible goods from authorship of the ideas or

expression embodied in them. The decision is not permission to make every deceptive affiliation claim. [53]

The principle matters beyond its precise US doctrinal setting. Different legal instruments have different jobs. A business should not assume that losing a copyright argument automatically produces a trademark argument with the same desired scope. Doing so risks turning a potentially renewable source-identification right into indefinite control over cultural content.

Consider four versions of Adrian's rival product. In the first, the competitor uses a clearly different identity and independently implements a public functional specification. In the second, it copies protected source code while keeping a different identity. In the third, it writes independent code but presents the product as Adrian's official release. In the fourth, it both copies code and impersonates the source. These are not four degrees of the same wrong. They are different combinations of potential copyright and trademark issues, with other rights remaining possible.

Mara faces an equivalent matrix. A new explanation of the same scientific topic under an independent title may be legitimate competition. A disguised adaptation of her distinctive expressive structure may require copyright analysis. A fake official edition adds source confusion. A review with clear attribution may involve permissible discussion. Classifying every version as "stealing my book" prevents a proportionate response because it erases the reasons why one version differs from another.

This distinction also disciplines enforcement. A complaint should identify the particular protected interest, the act, the relevant evidence, and the remedy requested. A seller who has copied a logo should not automatically be ordered to abandon all lawful functional competition. A rival who has copied protected text should not be allowed to dismiss the issue merely because its own company name appears on the cover. Precision protects both the claimant and the person accused.

There is also an ethical limit to the strategy of competing through brand. The author's name should represent genuine work, not merely become a premium label attached to unreviewed machine output. The software brand should identify a real maintenance obligation, not a marketing shell around indistinguishable and unaudited components. Where the institution cannot explain what it contributes, trust becomes a mechanism for charging more without doing more.

The chapter's practical conclusion is therefore qualified. Trademarks can help preserve distinguishable responsibility in a market full of easy substitutes. They cannot preserve every old margin, prevent every independent explanation, or make an unreliable product reliable. Their value is strongest when attached to a verifiable promise: this edition was checked by these people; this release will be maintained by this organization; this error-reporting route leads to someone accountable.

Before moving to books, ask a final classroom question. If the competitor changes its name and removes every misleading claim of affiliation, what part of your complaint remains? Whatever remains may be important. But it will have to stand on its own legal and factual foundations. That is where the copyright analysis begins.

4 — The book that returns in different words

The disappearance of matching sentences does not tell us what survived.

4.1 — Five meanings of “equivalent”

Mara uploads two manuscripts to a comparison tool. It reports little textual overlap. She feels that the second book nevertheless reproduces the first. The publisher replies that the sentences are new. Both are pointing to a genuine feature of the evidence. Neither has yet identified the right level of comparison.

We need at least five meanings of equivalence. Lexical equivalence concerns words and strings. Expressive similarity concerns the creative form in which material is presented, potentially including more than literal wording. Semantic equivalence concerns what propositions or meanings a passage conveys. Functional equivalence concerns what a reader can accomplish after using it. Commercial substitutability concerns whether one product displaces demand for another. These are analytical distinctions for this course, not five statutory categories.

Two passages can be semantically similar and expressively different. Two manuals can enable the same task while using different examples and structures. Two novels can compete for the same evening’s reading without sharing any distinctive plot. Conversely, changing every sentence may leave a detailed creative sequence substantially intact. Copyright’s exclusion of ideas and methods, together with protection for qualifying expression and adaptations, is why the distinction matters. [1, 3, 8]

Consider a short teaching passage invented here: “A town trusted one bell to announce a flood. When the bell stayed silent, the council concluded that the river was safe. No one had checked whether the bell still

worked.” A second passage says: “The warning siren made no sound, so the villagers assumed there was no danger. They had forgotten to test its battery.” The words and objects differ, but the miniature scenario and instructional turn are close. That observation supports further examination; it is not an automatic judicial conclusion about a tiny hypothetical text.

A third passage says: “An absent warning supports a reassuring conclusion only when the warning mechanism is sufficiently reliable.” It conveys the general reasoning lesson without reproducing the little narrative. A fourth develops the lesson through an independent discussion of missing laboratory measurements. The progression illustrates why “same meaning” is too broad to define infringement and “different words” is too narrow to exclude it.

Now scale up from three sentences to a book. Suppose the competitor retains forty distinctive examples, their order, their surprising reversals, and the relationship between the examples and chapter endings. That combination deserves a different analysis from a standard progression through introductory definitions, basic examples, and advanced applications. Conventional teaching order may be constrained by the subject itself. Unusual creative organization may carry a different evidential and legal significance. The analysis must separate the two rather than call every structure proprietary.

A translation provides the simplest warning against word-count theories. A translated chapter can have almost no identical words while still being an adaptation requiring permission, absent an applicable exception or other legal basis. Copyright law has long addressed transformations across form and language. An LLM is a new mechanism for transformation, not the inventor of nonliteral copying. [2, 8]

The practical question is therefore not “How much must I change?” It is “What am I using, what is protected, and what authorizes this use?” A percentage target invites a cosmetic exercise designed around a detector rather than the law. The US Copyright Office expressly warns against fixed numerical rules for fair use. A short protected passage can

matter greatly; a long use of unprotected facts raises a different question. [57]

For a publisher, the five-way distinction becomes a useful review form. Record literal overlap separately from narrative or structural correspondence. Identify shared facts and necessary conventions separately from discretionary expression. Assess market substitution separately from the legal basis. The result will be more demanding than a single similarity score. It will also tell the decision-maker why a close match matters—or why an emotionally persuasive allegation may reach beyond the right actually held.

4.2 — Fiction, nonfiction, style, and plagiarism

A novelist and a textbook writer can use the same word, “copying,” while objecting to different things. The novelist may recognize a distinctive chain of events and relationships. The textbook writer may see a competitor repeat the central explanation of a public scientific principle. Both may have suffered competition. The legal question is whether protected expression, rather than merely subject matter or knowledge, was appropriated.

For fiction, the level of abstraction matters. “A stranger arrives in a divided town” is an extremely broad premise. A fully developed sequence of scenes, revelations, character interactions, and original dialogue is much more concrete. Genre conventions do not become one writer’s property simply because that writer used them effectively. Nor can a detailed adaptation necessarily be made independent by replacing names and moving the setting. These are applications of the distinction between ideas and expression, not a promise that every court draws the line at the same place. [1, 8]

Style is particularly slippery. Short sentences, dry irony, fragmentary narration, or a melancholy atmosphere describe broad expressive tendencies. A general request for such qualities does not identify a specific protected passage. But a system can produce actual protected expression while ostensibly being asked only for a style. The legal analysis

should inspect the output and its relationship to identifiable works, not treat the prompt's label as decisive. Separate identity and endorsement concerns may also arise when a real person's name is used in marketing.

Nonfiction raises another boundary. Facts do not become owned merely because uncovering them was expensive. In *Feist*, the US Supreme Court rejected labour alone as the basis of copyright and distinguished unprotected facts from original selection or arrangement. In *Baker v. Selden*, copyright in a book explaining an accounting system did not provide exclusive control over using the system. Together, these decisions expose why a useful book can create value that copyright does not reserve to its author. [54, 55]

Mara cannot own the principle that unreliable measurements require caution. She may nevertheless have rights in her particular text and other qualifying creative choices. A competing author who investigates the underlying subject, consults independent evidence, and creates new explanations is in a different position from someone who feeds Mara's chapter to a model and requests a disguised substitute. The latter process is evidence of dependence, although dependence alone still does not eliminate the need to identify protected material and assess any defence.

Plagiarism is a related but different concept. In academic practice, presenting another person's ideas or expression as one's own can violate attribution norms even when copyright does not prohibit the use. Conversely, a person can clearly credit an author and still reproduce protected material without permission or an applicable exception. Citation answers a question about acknowledgment. It does not automatically answer a question about authorization.

Consider a public-domain historical essay. Republishing it with honest identification may be legally permissible, while silently presenting it as newly written personal scholarship would be academically deceptive. Consider a recent novel. Printing a substantial unauthorized adaptation with a prominent thank-you does not become permitted merely because

the source is acknowledged. One example separates legality from intellectual honesty; the other separates intellectual honesty from legal permission.

Editorial integrity therefore requires two reviews. A rights review asks whether uses are authorized or justified. An attribution review asks whether the publication accurately represents intellectual debts and contributions. A third factual review asks whether the claims are true. None can safely substitute for the others. A properly licensed text can be false. A truthful text can be unlawfully reproduced. A noninfringing summary can misrepresent who developed the ideas.

There is a final trap in nonfiction: copying mistakes. Suppose Mara invents a city name in a fictional exercise, and the rival book repeats it as historical fact. The match may support an inference of dependence, but it is not a magical entitlement to all surrounding facts. It is also a warning about epistemic quality. A production process optimized to preserve useful-looking meaning can preserve errors with equal efficiency. The task is not merely to make the language independent; it is to make the reasoning and evidence genuinely accountable.

4.3 — Training, retrieval, rewriting, and publication are separate acts

“Was the book used by AI?” is too coarse a question for serious analysis. A file may be acquired, stored, cleaned, tokenized, included in a training dataset, placed in a retrieval index, inserted into a model’s context, summarized, adapted, or published. Different actors may perform different steps in different countries. The research framework developed by Lee, Cooper, and Grimmelmann usefully treats generative AI as a supply chain rather than a single indivisible act. [49]

Training generally changes model parameters through a learning process. Supplying a book in a prompt does not necessarily train the model in that sense. Retrieval-augmented generation, often abbreviated RAG, retrieves material from an external collection and supplies selected content to a model when answering. Fine-tuning performs an

additional learning process on selected data. These technical distinctions matter because the location, purpose, persistence, and exposure of copies can differ. A system’s marketing description may conceal those differences.

Imagine a publisher with permission to store a manuscript for ordinary editing. An employee uploads it to a third-party service. The service uses it only for that session in one configuration, but retains it for improvement in another. These are materially different data flows, even before assessing the contract and applicable law. A general statement that the employee “used an assistant” is not enough to decide which rights or confidentiality obligations were implicated.

The same separation works in reverse. A developer could have lawfully assembled training material and nevertheless operate a system that reproduces protected passages in problematic outputs. Another system might produce an independently worded answer while relying on unlawfully acquired source files. A low output-similarity score does not retrospectively authorize acquisition. A lawful input route does not prospectively authorize every possible output.

Memorization is not merely a philosophical possibility. Carlini and colleagues demonstrated extraction of verbatim training examples from GPT-2. Their work establishes a concrete risk under the studied conditions, not that every current model stores every training book as an easily recoverable archive. A responsible assessment must distinguish a demonstrated mechanism from an unsupported universal claim. [44]

The legal debate about training remains sensitive to facts and jurisdiction. The US Copyright Office’s May 2025 pre-publication report on training rejects simplistic treatment of all training as one kind of use. It discusses purpose, source acquisition, outputs, market effects, and licensing. At the research cutoff, the Office’s AI page still identified Part 3 as a pre-publication version. It is an official policy analysis, not a statute or a binding resolution of every pending case. [15, 16]

Litigation developments require equally careful labeling. On 20 July 2026, the district court granted final approval of the class settlement in

Bartz v. Anthropic, involving a \$1.5 billion settlement fund and disputed downloading of books from pirate libraries. The order also addresses destruction of the relevant original files and copies, subject to preservation obligations. This is a substantial, concrete resolution of a particular dispute. It is not a Supreme Court declaration that all AI training is lawful or that all AI training is infringement. [48]

For EU projects, the distinction between the research-oriented and broader text-and-data-mining exceptions described in Chapter 2 remains important. Lawful access, purpose, eligible institutions, rights reservations, and the particular act must be examined. Separately, the AI Act imposes obligations on general-purpose model providers concerning copyright policy and training-content summaries. Those regulatory duties do not themselves decide whether a specific generated book infringes copyright. [21, 22]

The recommended operational response is a data-flow map with rights decisions attached. For each stage, record the material, purpose, actor, location, retention, access conditions, and intended downstream use. Such a map does not guarantee legality. It creates an object that lawyers, engineers, editors, and auditors can actually evaluate. Without it, organizations tend to argue about the word “AI” while remaining unable to say where the manuscript went.

4.4 — Who authored the AI-assisted book?

Mara experiments with three workflows. In the first, she writes a chapter and uses an assistant to flag awkward sentences. In the second, she creates a detailed argument, selects evidence, drafts examples, and rewrites machine-proposed passages into her own expression. In the third, she asks for an entire book and accepts the result with minor corrections. The fact that all three involve a human and a model does not make the human contribution identical.

The US Copyright Office’s January 2025 report explains that human-created expression can remain protected in AI-assisted works, while purely generated material or material without sufficient human

expressive control does not acquire protection on that basis. Creative selection, arrangement, and modification can matter. Prompts alone generally do not supply the required control under the systems examined. The assessment is case-specific and concerns expression, not a tally of prompts or hours spent. [14]

The useful question is therefore evidential: what did the human actually create that is perceptible in the result? A researcher may have contributed an original conceptual framework; a writer may have composed the text expressing it; an editor may have creatively reorganized material; a machine may have supplied most of the wording. Those contributions can coexist without merging into a claim that one person necessarily authored every protected element.

A purchase contract with an AI provider does not settle this threshold question. Contractual language can allocate whatever rights the contracting parties possess or promise cooperation concerning them. It cannot force a jurisdiction to recognize copyright in material that fails the legal requirements for protection. Nor can an output-ownership clause extinguish third-party rights that the provider never owned.

National differences remain relevant. The United Kingdom has a statutory framework addressing certain computer-generated works and was considering reform in its March 2026 policy report. That report should not be read as if proposed removal or amendment had already occurred. A publisher distributing internationally should avoid reducing all authorship questions to one US office statement, while still recognizing its importance for US registration and practice. [34]

For evidence, Mara keeps substantive drafts, source notes, revision histories, and a record of important editorial decisions. She need not preserve every keystroke merely to look human. The objective is to show real contributions and permissions, not to manufacture an artificial performance of effort. A prompt log can help reconstruct a process, but quantity is not quality: a thousand instructions to “try again” do not necessarily establish authorship of the final expression.

The more uncomfortable question concerns editorial responsibility. Even where a publisher cannot claim copyright in every generated sentence, it can still be responsible for choosing to distribute the book. It can investigate factual claims, check citations, remove misleading attributions, obtain necessary permissions, and correct errors. Those obligations are not made less important by uncertainty about exclusive rights.

This book applies the same principle to itself. Its front matter discloses AI-assisted drafting and does not invent a human author for all of its expression. Its educational value should be assessed through the arguments, examples, and sources, not through a fictional origin story. Substantive human review before publication remains important. A book explaining transparency should not conceal its own production conditions behind a decorative copyright notice.

The constructive lesson is to design the creative process around genuine intellectual work rather than legal cosmetics. Use assistance where it helps. Preserve human decisions where they matter. Describe the result honestly. Rights can then be assessed against an actual creative record, while readers receive a more accurate account of who stands behind the publication.

4.5 — When the substitute is lawful but the damage is real

Return to the most difficult version of Mara’s problem. The competitor consults lawful sources, does not reproduce protected expression, uses an independent brand, and creates a genuinely different explanation. Its book nevertheless gives many readers enough of what they wanted from hers. Assume, for this thought experiment, that no other legal prohibition applies. Mara loses sales. What exactly should copyright do now?

The answer cannot be inferred from the loss alone. Markets regularly contain lawful substitutes: a lecture instead of a textbook, a new textbook instead of an old one, a library loan instead of a purchase, or an

open tutorial instead of a paid course. A rule prohibiting every effective substitute would reserve not merely an expression but much of its subject matter and practical purpose. The legal system's distinction between ideas and expression would become hollow.

AI may make such substitution cheaper and more individualized in some settings. The economic concern is not necessarily that every model can reproduce every valuable book. It is that a reader may accept a lower-cost answer, tailored explanation, or compressed guide that is good enough for a particular purpose. The original publisher then competes with something that need not match the book's full quality to reduce demand. This is a scenario analysis, not a measured prediction of the entire publishing market.

An invented example clarifies the asymmetry. Suppose producing and checking a specialist book costs €40,000. A competing explanation costs €1,000 to generate and package, but has not undergone comparable verification. These figures are assumptions. If readers cannot distinguish reliability before purchase, the cheaper product may exert price pressure disproportionate to its quality. Alternatively, readers may value the original author's care and choose the verified edition. The result depends on institutions of discovery, evaluation, and trust, not on generation cost alone.

The most important productive response may therefore lie outside the text itself. A verified edition can offer traceable references, corrections, exercises with carefully checked answers, access to the author's evolving analysis, and a coherent learning path. Institutions can purchase editorial reliability for students or staff. None guarantees income. Each makes the creator's contribution more observable instead of pretending that useful knowledge can be sealed forever inside one wording.

There is also a collective-action problem. If new explanatory systems depend on a continuing supply of expensive reporting, research, and careful synthesis, a market that pays only for cheap repackaging could underfinance the upstream work. Whether this happens, and to what extent, is an empirical question. It should be investigated through

production incentives, entry, diversity, quality, and long-term investment—not asserted from one dramatic example.

Policy responses might include targeted licensing where rights apply, collective arrangements with appropriate governance, public or philanthropic funding for socially valuable works, institutional subscriptions, and support for open educational resources. These mechanisms do different things. Licensing compensates authorized uses of rights. Public funding can support work whose social value exceeds what individual readers will pay. Neither should be advertised as a universal cure.

Mara therefore needs two diagnoses. The first asks whether a competitor unlawfully used her protected expression or misled readers. The second asks whether her business remains sustainable under lawful substitution. Litigation may address the first and fail to address the second. Commercial adaptation may address the second without excusing the first. Keeping the diagnoses separate is not indifference to creators. It is a prerequisite for choosing a remedy that could actually work.

5 — The program that does the same thing

*A different implementation can be a legitimate competitor.
A renamed implementation can still be a copy.*

5.1 — Software has more than one identity

Adrian compares two screens. Both applications import research papers, connect claims to sources, and export an evidence report. The resemblance is commercially significant. It tells him almost nothing, by itself, about whether the underlying source code was copied. To investigate that question, he must separate the program's function from its implementation, documentation, interface assets, data, and identity.

Internationally, TRIPS treats computer programs in source or object code as literary works. The EU Software Directive protects the expression of a computer program while excluding ideas and principles underlying its elements, including interfaces. These rules do not turn every useful algorithm or capability into copyright property. They also do not reduce software protection to identical strings alone. [3, 20]

An implementation contains choices at several levels. There are individual expressions and statements, functions and modules, internal structures, external interfaces, and larger organizational patterns. Some choices reflect creative discretion. Others follow from an algorithm, efficiency, a platform, a standard, or compatibility requirements. Before comparing two programs legally, an analyst must understand which is which.

Computer Associates v. Altai developed the influential US abstraction–filtration–comparison approach. In simplified terms, the analysis moves through levels of program structure, filters out elements that should not receive copyright protection, and compares the remaining protectable material. It is a judicial method developed in a particular

legal setting, not a universal algorithm that automatically decides all software cases. Its lasting educational value is the insistence on filtering before treating resemblance as wrongful appropriation. [56]

Consider a function that sorts records by date. Both programs may use a familiar algorithm and expose an input parameter called “date.” Those similarities can arise from the task rather than copying of protected expression. Now suppose both contain a long unusual comment, a distinctive arrangement of unnecessary helper functions, and the same idiosyncratic error message. The second set of correspondences raises different evidential questions. Neither set should be judged without technical context.

A modern application also contains assets that do not become unprotected merely because they are packaged with code. An original illustration, a substantial manual, an explanatory video, or a distinctive set of written tutorials requires its own analysis. A developer may be free to implement a function yet not free to copy the manual that explains it. Reimplementing the engine does not clear every component of the product.

Patents and trade secrets occupy another layer. Independent development is not a general defence to patent infringement where a valid applicable patent covers the implementation. Trade-secret law can prohibit improper acquisition or use of confidential material, while preserving routes such as independent discovery and certain lawful observation or testing under the relevant rules. Contractual restrictions can add further questions. Copyright’s limits therefore do not establish a universal freedom-to-operate opinion. [27, 58]

Adrian’s first useful document is an asset map. It identifies source code, third-party libraries, user-interface artwork, documentation, datasets, brands, confidential know-how, and any relevant patent interests. Each asset has a source, an owner or claimant, permissions, and a proposed use. This map is not bureaucracy for its own sake. Without it, a team may spend weeks comparing source code while overlooking a copied tutorial or an unauthorized logo.

The conceptual lesson is that a software product is not a single legal object. It is a bundle of interacting objects, behaviours, and relationships. AI-generated code does not collapse those layers into one new category. It makes good separation more important because a model can mix conventions, remembered fragments, new expression, and third-party identifiers in one apparently coherent answer.

5.2 — Interoperability, APIs, and the limits of reimplementation

Imagine that Adrian’s application is the only tool that can read a customer’s archive. The customer wants a competing tool to import the same files. Should Adrian’s copyright necessarily entitle him to prevent that? The question concerns not only investment in software but also control over the user’s accumulated work. Interoperability can therefore produce a conflict between an incumbent’s interests and the public value of competition and portability.

In *SAS Institute v. World Programming*, the Court of Justice of the European Union held that a program’s functionality, programming language, and data-file formats used to exploit functions were not forms of expression protected as computer programs under the Software Directive. The Court’s official summary also emphasizes lawful observation and study, while preserving the possibility of infringement through protected material from a manual. The facts included the absence of access to the original source code. [19]

This is not a licence to copy every file format specification verbatim, misuse confidential material, or ignore other rights. It illustrates a narrower principle: protecting the expression of a program is not the same as controlling every functional means by which another program can work with it. The exact source of information and the implementation process remain important.

The US Supreme Court’s *Google v. Oracle* decision reached a different legal question through fair use. Assuming without deciding that the relevant material was copyrightable, the Court found the particular

reuse of Java API declaring code at issue to be fair use. The decision concerned approximately 11,500 lines used in a new computing environment. It did not declare all APIs everywhere uncopyrightable or authorize wholesale copying of arbitrary implementation code. [18]

The distinction between these cases is educational. A result can follow because something is outside the protected subject matter, because the allegedly copied elements are unprotected, or because an exception permits a use even though protected material is involved. These routes should not be merged into the slogan “interfaces are free.” Their conditions and consequences differ.

EU software law also distinguishes observation or testing during authorized acts from decompilation for interoperability. The latter exception has conditions, including indispensability and limits on the information obtained and its use. It is not an unrestricted right to reconstruct a competitor’s product for any purpose. The enforceability of conflicting contractual provisions also requires attention to the Directive’s mandatory rules. [20]

For Adrian, the practical question becomes how to document a legitimate implementation process. A lawful public specification may provide an adequate starting point. Authorized testing can identify input-output behaviour. A team can write its own implementation and documentation, record the reasons for compatibility choices, and avoid importing proprietary source or confidential notes. This is a proposed evidence-conscious process, not a statutory safe harbour.

A clean-room arrangement separates the people examining permitted reference material from those implementing an independently expressed specification. Its value lies in reducing exposure and clarifying provenance. It does not legalize an unlawfully obtained specification or neutralize patents, contractual restrictions, or protected expressive material passed through the supposed barrier. An LLM that has seen unknown training data adds uncertainty rather than making the barrier perfect.

One especially weak practice is to place proprietary source code into a model and ask it to rewrite the code “so that it is legally independent.” A request of that form demonstrates neither independence nor permission. The resulting program may preserve protected structure, may be an unauthorized transformation, and may expose confidential material to a service. Changing the tool used for copying does not answer the legal question.

A stronger engineering aim is narrower and testable: implement a defined lawful interface with independently developed code, preserve evidence of permitted inputs, and review actual correspondences before release. That approach accepts uncertainty without surrendering legitimate interoperability. The point is not to make competition impossible to challenge. It is to make the reasons supporting lawful competition intelligible.

5.3 — Open source is permission with terms, not abandoned ownership

A coding agent suggests a useful function. It runs. The team copies it into its repository and moves on. Later, someone finds a similar implementation in a public project. At this stage, two bad reactions are common: assuming that everything public is unrestricted, or assuming that any resemblance makes the entire company subject to an incompatible licence. Neither reaction begins with the facts.

Open-source licensing generally operates through permissions granted by rights holders, subject to the relevant licence’s conditions. Public availability alone does not provide those permissions. Conversely, a valid open-source licence can authorize substantial uses, including commercial uses, that would otherwise require individual negotiation. The actual text and the manner of use matter more than the informal label “open.” [40, 41, 42, 43]

The MIT licence grants broad permissions while requiring preservation of the copyright and permission notices in copies or substantial portions. Its warranty disclaimer is also important. Apache License 2.0

adds more detailed provisions, including conditions concerning modifications and notices, an express patent licence with defined scope and termination conditions, and a limitation concerning trademarks. These are primary licence texts, not evidence that every project using their names has a clean ownership history. [40, 41]

GPL version 3 takes a different approach to certain covered distributions or conveyances: the relevant permissions are tied to copyleft requirements, including provision of corresponding source in the circumstances defined by the licence. This is not a rule that learning a technique from GPL software makes every later implementation GPL. The protected material used, whether a covered or derivative work is involved, and the mode of distribution need examination. [42]

AGPL version 3 adds a network-interaction provision for modified versions of the covered program. Describing this as “all SaaS must reveal its code” is inaccurate. The question is whether the licence’s actual conditions apply to the particular covered work and modification. A remote interaction with an unrelated service does not magically expand the licence to every program in the organization. [43]

The lesson for LLM-generated code is neither panic nor licence laundering. A model’s removal of a header does not remove obligations attached to protected material that has been copied or adapted. But independently implementing an unprotected function does not acquire another implementation’s licence merely because the two solve the same problem. Establish the relationship before choosing the compliance response.

A useful review begins with a suspected fragment. Locate plausible sources and inspect their licences at the relevant version. Determine whether the similarity concerns banal syntax, a standard algorithm, or a more distinctive implementation. Check what the project actually incorporated and distributed. Escalate ambiguous cases instead of either concealing them or announcing that the whole repository is contaminated. Replacement with genuinely independent code can be appropriate, but a cosmetic second rewrite is not automatically independent.

Licence compatibility also matters when combining components. A permissive label is not a guarantee that all obligations disappear; a copyleft label is not an explanation of every boundary case. Organizations should maintain an approved component policy, a review route for unfamiliar licences, and accurate notices for the release being shipped. The policy should be practical enough that developers can follow it without inventing informal exceptions under deadline pressure.

Finally, copyright permission and trademark permission should remain separate. An open-source project may authorize modification of its code while retaining control over the source-identifying name or logo. A fork may need a new identity to avoid suggesting official maintenance. This is not a contradiction. It permits technical freedom while helping users identify who is responsible for a particular distribution. [40]

For Adrian, good open-source practice can be a competitive advantage. Customers may value a company that knows what it ships, can produce its notices, and can explain the provenance of critical components. That confidence comes from a maintained process, not from claiming that the AI provider made all licensing questions disappear.

5.4 — Testing equivalence is not testing independence

Adrian commissions two evaluations of the rival program. One asks whether it behaves like his application. The other asks whether it reproduces protected code. It would be convenient if one test answered both questions. It does not. Behavioural equivalence concerns what programs do; copyright analysis concerns the protected material and legally relevant acts. A program can behave differently because it is a buggy copy. Another can behave identically on a defined task because it is independently implemented.

Even behavioural equivalence requires a specification. Does “same” include error messages, timing, memory use, ordering of equal records, treatment of malformed input, network effects, and persistence after failure? Two functions can agree on normal examples and diverge at a

boundary. A useful comparison states which observations matter, over which input domain, and under which environmental assumptions.

Suppose both applications sort evidence records by date. The test suite contains one hundred well-formed records with distinct dates. Both pass. The suite says nothing about duplicate dates, missing dates, time zones, extremely large files, or corrupted encodings. This is not a criticism of testing; it is an explanation of what the test result establishes. A statement such as “equivalent for the tested cases” is evidence. “Equivalent in all circumstances” requires much more.

There cannot be a general terminating procedure that decides complete behavioural equivalence for every arbitrary program in a sufficiently expressive programming system. A brief reasoning argument shows why. Compare a program that immediately returns zero with one that first simulates an arbitrary computation and returns zero only if that computation halts. A perfect equivalence decider would tell us whether the simulated computation halts. This connects the unrestricted equivalence problem to the halting problem. Restricted languages, bounded domains, and specified properties can still admit useful verification.

Legal independence is harder to mechanize in a different way. Textual matching can reveal literal fragments. Structural comparison can reveal correspondence after renaming or reordering. Neither tool decides whether the matching material is protected, whether copying occurred, whether permission exists, or whether an exception applies. A detection system should therefore produce evidence with explanations rather than an unsupported binary verdict.

A proposed review pipeline starts with exact and approximate matches, then groups candidates by technical context. A common generated boilerplate function should not receive the same risk score as a distinctive copied comment and implementation sequence. Reviewers record the source, licence, similarity, functional constraints, and disposition. High-risk unresolved components can be quarantined before release. The pipeline improves consistency; it does not eliminate the need for judgment.

A software bill of materials helps answer a different question: what components are present? SPDX provides standardized ways to represent information about software components and licensing. Such metadata supports inventory and exchange, but it does not prove that every ownership assertion is true or every missing component has been identified. A beautifully formatted inventory can still be incomplete. [52]

Signed commits and release signatures can help establish integrity and identify the key used to approve a version. They do not, by themselves, prove that the signer authored the code or had authority to license it. A reliable record should connect technical artifacts with contributor agreements, review decisions, and actual organizational responsibilities. Cryptography can protect the record from certain alterations; it cannot fill an empty evidential field with truth.

The final separation is between legal and engineering quality. A fully licensed program can contain a severe security defect. An independently implemented program can corrupt data. A technically excellent program can carry unresolved licence obligations. Adrian needs release gates for all three questions: rights, behaviour, and security. Allowing one successful test to stand in for the others is an attractive shortcut precisely because the missing failures are initially invisible.

5.5 — What remains valuable when code becomes cheaper?

Assume that AI assistance substantially reduces the cost of producing a first implementation of some feature. That assumption does not establish that the entire product has become cheap. A feature must fit requirements, integrate with existing systems, survive unusual inputs, protect data, remain maintainable, and be corrected when conditions change. For some products those costs may dominate; for others a cheap substitute may be enough. The responsible conclusion is differentiated rather than triumphalist.

Adrian should therefore ask what customers actually buy. They may buy code access, but they may also buy operational reliability, migration help, support, accountable handling of incidents, documentation, integrations, or a continuing compatibility promise. Some of these contributions are protectable assets; others are services or organizational capabilities. They should not be collapsed into the phrase “our intellectual property.”

A hypothetical customer comparison makes the point. Product A is cheaper at launch but has no tested recovery procedure. Product B is more expensive but comes with maintained exports, reproducible releases, and a team responsible for restoring service. A customer performing a trivial personal task may rationally choose A. A laboratory whose evidence archive supports years of work may rationally choose B. Neither choice proves that copyright is obsolete. It shows that different users value different bundles.

The maintenance argument should not become an excuse for unnecessary dependence. A supplier can create apparent value by making migration painful, concealing file formats, or attaching obscure restrictions to ordinary data access. Some such arrangements may raise separate legal questions; even when permitted, they deserve ethical scrutiny. A business model based on customers’ inability to leave is different from one based on customers’ confidence that staying is worthwhile.

Open standards and interoperability can support another equilibrium. Several firms may implement the same basic capability while competing on verification, support, and specialized features. Shared components can reduce duplicated effort. The corresponding danger is underfunding common infrastructure if every beneficiary expects someone else to maintain it. Financing mechanisms—support contracts, shared foundations, member contributions, or institutional procurement—should be assessed against that problem rather than assumed to arise automatically.

Adrian’s practical plan can therefore combine selective protection and openness. He may keep some legally protectable code proprietary,

publish a stable interchange format, use open-source components with proper compliance, and offer paid support. Another company may choose a fully open model. These are strategic options, not universal prescriptions. Their viability depends on the product, customer base, maintenance burden, and incentives of contributors.

When facing a suspected clone, he should pursue a proportionate investigation. Preserve the released versions and public claims. Identify the specific code or other assets allegedly copied. Distinguish functional similarity from expressive correspondence. Check his own chain of title before asserting rights. Review possible permissions and independent-development explanations. Then choose a response: technical clarification, a negotiated licence, correction of branding, replacement of a component, or formal proceedings where justified.

The important discovery is that code generation and software production are not identical activities. Copyright can still matter greatly to code and documentation, but it cannot guarantee a monopoly over a useful result. A resilient software organization knows which part of its advantage rests on exclusive rights and which part rests on doing difficult work well. AI makes that accounting more urgent; it does not perform the accounting on the organization's behalf.

6 — Solutions without owning all meaning

A useful solution must say which problem it solves—and which problem remains.

6.1 — Start with a rights ledger, not a prohibition banner

Mara considers adding “No AI use of any kind” to every page. Adrian considers a policy banning all coding assistants. Either measure might express a preference. Neither, by itself, tells the organization what it owns, what others may lawfully do, or what employees actually need to accomplish. The first practical solution is less dramatic: inventory the rights and the uses before deciding where to impose controls.

A rights ledger is a proposed operational record, not a new statutory register. For each asset, it identifies the version, creator or source, claimed owner, supporting agreement, permitted uses, relevant territories, restrictions, expiry or review date, and unresolved questions. The ledger distinguishes verified ownership from a contributor’s unsupported assertion. It also records when a work is used under an exception rather than a licence. A green “cleared” label without scope is not enough.

For Mara’s book, one entry might cover her manuscript, another a commissioned cover, another a licensed photograph, and another an extract used for criticism. The entries should not imply that permission for the photograph covers the photographer’s subject’s identity rights, or that an ebook licence necessarily covers a translated audiobook. Separate assets and separate uses require separate reasoning. The ledger makes those distinctions visible before a new format is released.

For Adrian, the same structure can cover employee code, contractor contributions, dependencies, documentation, datasets, and trademarks. It can record the evidence supporting the company’s entitlement to

release each component. The implementation need not be elaborate: a small team can begin with a controlled document or repository file. The quality lies in the accuracy and maintenance of the information, not in the sophistication of the database.

An AI-processing agreement should describe activities rather than rely only on a sweeping word such as “use.” Internal editing, temporary retrieval, fine-tuning, general model training, creation of derivative editions, public outputs, and sublicensing are materially different. The parties can negotiate retention, deletion, access controls, permitted providers, reporting, audit, and responsibility for claims. Whether particular terms are enforceable remains a question of applicable law and mandatory limits.

Risk allocation should also be readable. An indemnity is a contractual allocation of specified losses; it is not proof that a use is lawful. Its value depends on coverage, exclusions, procedure, limits, and the counterparty’s ability to perform. A small publisher should not assume that a reassuring sentence in a service description will finance every dispute about a generated book. Likewise, a developer should not promise customers more protection than its upstream contracts and actual process support.

The release decision can use three internal states. “Approved” means approved for a stated use on stated evidence. “Conditional” means specified obligations remain, such as notices or a restricted distribution route. “Unresolved” means the material cannot pass the relevant release gate without further review. These are proposed management categories, not legal determinations. Their advantage is that uncertainty becomes work to complete rather than a vague anxiety passed to the next team.

A final record should identify the accountable person and review trigger. A change in model configuration, a new distribution territory, a newly discovered source match, or a new edition may require reassessment. This prevents yesterday’s permission from silently becoming a perpetual approval for every future product. Good governance is not the

elimination of uncertainty. It is the ability to say where uncertainty remains and who must act when the facts change.

6.2 — Provenance, signatures, and the limits of technical protection

A hash can answer a narrow question extremely well: does this file match the bytes represented by this digest, assuming the chosen cryptographic function behaves as intended? It cannot answer who created the file or who owns its copyright. A person can hash someone else's manuscript. A timestamp can support a claim that particular data existed by a certain time, within its trust assumptions; it does not prove original authorship. The same limits persist when the record is placed on a blockchain.

This is not an argument against cryptography. It is an argument for assigning it the right job. Integrity, authenticated assertions, and durable chronology are useful components of an evidence system. Legal ownership requires additional facts about creation, employment, transfers, licences, and applicable law. A mathematically valid signature on a false ownership assertion remains a valid signature on a false assertion.

C2PA provides a technical architecture for cryptographically verifiable provenance information. Its trust model concerns signed claims and the identity or credentials associated with a signer; a consumer still needs to evaluate the assertions. This book cites version 2.2 as a concrete specification, not as a claim that it is the newest release or a universal copyright registry. Provenance can improve interpretation without certifying that the underlying content is true or lawfully owned. [46]

For Mara, useful provenance might connect a released edition to its editorial history and approved publisher. For Adrian, it might connect a release to source revisions, component inventories, and review approvals. These records should be designed for a real dispute: can the organization explain the chain, reproduce the relevant version, and show

which claims were independently verified? A decorative authenticity badge with no accessible evidence is much weaker.

Rights-reservation signals solve a different problem. The W3C-hosted TDM Reservation Protocol community has developed machine-readable mechanisms intended to communicate reservations and licensing routes. Such signals can help automated actors discover an expressed position. Community specifications are not themselves legislation, and a signal's legal effect depends on the relevant rule and circumstances. It is not a universal ban on every analytical activity. [47]

The Robots Exclusion Protocol makes another boundary explicit: crawler instructions are not access authorization or a security mechanism. Publishing a robots rule can communicate a preference to cooperative crawlers, but it does not prevent a determined person from requesting a public resource. Conversely, the absence of a robots prohibition does not establish that all copyright uses are licensed. Security controls, legal permissions, and machine-readable preferences are separate layers. [45]

Watermarking and fingerprinting should be assessed against a stated threat model. Exact fingerprints are useful for exact copies. They fail by design when different bytes are compared. More tolerant matching can detect some transformations but may introduce false positives. A watermark intended to survive paraphrase needs evidence that it does so under the relevant transformations; its mere presence in a product description proves little. No particular watermark is endorsed here as a general solution to semantic rewriting.

Monitoring also has costs. Scanning private manuscripts or proprietary repositories through an external service can expose sensitive material. Keeping every prompt indefinitely can increase confidentiality and privacy risk. A sensible design minimizes unnecessary collection, restricts access, separates public provenance from confidential evidence, and records retention decisions. The aim is proportionate accountability, not universal surveillance of creative work.

The practical package is therefore layered: clear notices, appropriate reservation signals, controlled access where warranted, versioned evidence, limited matching tools, reviewable claims, and an accountable response process. No layer turns an idea into property. Together, they can make unauthorized copying harder to hide and authorized collaboration easier to document without pretending that a cryptographic primitive has solved a legal question.

6.3 — Licensing and compensation: useful mechanisms, hard allocation problems

A licensing market is attractive because it can replace uncertain confrontation with permission and payment. A publisher grants specified uses; a model developer obtains a defined corpus; both record the scope. The basic mechanism is familiar. The difficult questions concern who can license which rights, what uses require permission, how prices are negotiated, and whether small creators can participate on workable terms.

Direct licensing may suit a distinctive catalogue with clear ownership and substantial demand. It can also impose transaction costs that make very small deals impractical. Collective arrangements may reduce those costs by representing many rights holders. Yet aggregation creates governance questions: membership, representation, repertoire accuracy, withdrawal, distribution rules, and accountability. These mechanisms should be evaluated as institutions, not applauded merely because they contain the word “compensation.”

A creator may value different permissions differently. Training a general model, operating a controlled educational search service, creating an authorized adaptation, and selling excerpts can affect the creator’s plans in different ways. A single undifferentiated price may conceal these differences. A well-designed negotiation separates purposes and material, while respecting uses that do not require a licence under applicable law.

Compensation for training inputs also faces an attribution problem. Suppose a model produces an explanation of a familiar scientific principle after learning from thousands of overlapping sources. Which source contributed how much? Similarity to an output is not necessarily causal contribution to learning. Causal contribution is not necessarily legal entitlement. Legal entitlement is not necessarily a complete account of what society ought to reward. These are three different allocation questions.

Wang and colleagues propose a contribution-based compensation framework using the probabilistic structure of generative models and cooperative-game-theoretic ideas. This is a research proposal, not an enacted royalty system or a demonstrated universal solution for allocating payments across the world's books and software. Its importance lies in making the allocation question explicit and amenable to technical investigation. [50]

Several conceptual difficulties remain. A foundational source may influence many later works while being rarely reflected in literal output. Two sources may provide nearly interchangeable information. A low-quality source may increase a model's tendency to reproduce an error. A contributor may inflate a corpus with duplicates to seek additional payment. Any mechanism should be tested for such incentives rather than assume that a plausible formula will produce fairness by itself.

There is also a bargaining-power issue. A market in which a few major catalogues can negotiate and everyone else faces opaque take-it-or-leave-it terms may improve compensation unevenly. Conversely, a licensing rule so expensive or administratively complex that only the largest developers can comply may strengthen incumbents. A balanced design should measure transaction costs, access for smaller actors, and the concentration effects of its compliance requirements.

Open licensing offers another route when creators deliberately choose broad reuse. Creative Commons Attribution 4.0 permits uses under its conditions, including attribution and indication of changes where required. It does not license every separate right associated with

a work or permit a licensor to grant rights it does not hold. Its permissions cannot simply be withdrawn from compliant recipients because a later use becomes commercially inconvenient. [51]

A creator can therefore choose different models for different assets: open foundational material, paid verified editions, controlled licences for selected adaptations, or institutional funding for public access. That portfolio may serve some projects well and others poorly. The important point is informed choice. Compensation policy should not assume that every creator wants maximum exclusivity, while open-culture policy should not assume that every creator can finance substantial work without income.

6.4 — Enforcement that protects creators without automating censorship

Mara finds a suspicious book. The first impulse is to demand immediate removal from every platform. Sometimes a rapid response is justified, especially where the evidence is clear and harm is ongoing. But a reliable process must distinguish an exact unauthorized copy from a lawful review, an independent explanation, or a misidentified public-domain edition. A system designed only for speed can amplify the claimant's mistake.

A practical evidence packet should identify the claimant and basis of authority, the original work and relevant version, the allegedly infringing material, specific correspondences, and the requested remedy. It should state why the correspondences concern protected expression rather than merely a shared topic. For a trademark complaint, it should explain the protected sign and the misleading presentation. These are proposed evidential standards for organizational practice, not a substitute for the formal requirements of a particular procedure.

US law includes the section 512 notice-and-counter-notice framework and related limitations on liability for qualifying service providers. The statute also addresses knowing material misrepresentations in the relevant notices. This is not a worldwide automatic-removal

rule, nor does a platform's takedown decision conclusively determine infringement. Users and claimants must consider the actual procedure and consequences before acting. [12]

EU enforcement rules likewise require attention to effectiveness, fairness, and proportionality. A remedy should address the established problem without casually eliminating legitimate trade or imposing unnecessary barriers. The Enforcement Directive supplies a framework that operates through national implementation and procedure; it is not a universal litigation script. [28]

Automation can assist by organizing evidence, locating copies, clustering notices, and checking whether required fields are present. It should not be entrusted with an unexplained final determination that all semantically similar text is unlawful. A summary of a scientific principle may resemble many sources because they describe the same principle. A model trained to find "equivalents" may be excellent at detecting the wrong kind of similarity for a legal claim.

False positives have a distributional cost. A large publisher may be able to challenge a mistaken removal. A small educator may abandon the work because contesting the decision takes too much time. False negatives have an equally real cost for creators whose work is copied at scale. The solution is not to pretend that one error type is harmless. It is to design review, explanation, appeal, and escalation around both.

The remedy can also be narrower than deletion. An incorrect affiliation claim may require correction. A missing licence notice may require compliance and legal review of any other consequences. A copied module may be replaced or licensed. A disputed excerpt may justify negotiation over its scope. Some cases will still require injunctions or damages. Precision increases the chance that the response solves the actual problem instead of merely punishing the opposing party.

For Adrian, a sensible escalation ladder begins with preservation and technical assessment, then moves to a clear private inquiry where appropriate, qualified legal review, and formal action if justified. He should avoid public accusations unsupported by evidence. He should

also check his own rights before demanding exclusivity over a third-party component or a public functional convention.

The institutional goal is contestable enforcement: a person affected by a decision should be able to understand its basis and present relevant contrary evidence. This is a recommendation for responsible systems, not a statement that every platform presently offers such a process. It protects creators by improving the quality of their claims and protects users by preventing intellectual property from becoming an opaque instrument of removal.

6.5 — What policy should protect, and what it should leave free

The strongest case for intervention is not “a creator lost revenue,” but a more specific account of the behaviour, the right, the incentive, and the public consequence. Unauthorized copying of protected expression, misuse of confidential material, and deceptive branding can justify responses under existing legal frameworks. Lawful learning and independent implementation require different treatment. A policy that refuses this distinction risks punishing the very activity that copyright leaves available.

This book’s normative position is against a general monopoly over semantic equivalence. Such a monopoly would be difficult to define and potentially broad enough to control explanations, translations of ideas into practice, independent tutorials, and compatible software. It could allow the first persuasive account of a subject to constrain later accounts of the subject itself. A rule intended to protect creators could become a tax on understanding.

That position does not imply that all training or all rewriting must be free. Where protected acts require permission and no exception applies, authorization remains necessary. Where an adaptation preserves protected expression, changing words is not a complete answer. Where output impersonates a source, trademark and other rules may matter.

Defending the boundary around rights requires taking the rights inside that boundary seriously.

The more promising reforms are narrower. Improve access to relevant evidence without requiring public disclosure of every trade secret or private manuscript. Support interoperable rights information without converting optional metadata into a new formality for copyright existence. Facilitate licensing where rights apply while preserving research and lawful competition. Make remedies usable for smaller creators without depriving smaller defendants of a meaningful chance to respond.

Public financing belongs in the conversation because some valuable production may not be well funded by exclusive sales alone. Libraries, universities, professional institutions, public grants, and philanthropic organizations can support research, translation, preservation, and educational work. The design questions concern independence, selection quality, accountability, diversity, and capture. Public support is not automatically wise; private exclusivity is not automatically sufficient.

Any proposed compulsory or extended licensing regime would require explicit legal design, not a technical platform declaring itself authorized. It would need to address scope, eligibility, rates, governance, distribution, exceptions, oversight, and international obligations. A contribution-scoring system cannot enact those choices. At most, it can inform one part of the administration after legitimate rules exist.

Evaluation should focus on outcomes rather than the number of documents generated or licences signed. Are high-quality new works still being produced? Can new entrants compete? Are small creators receiving usable compensation? Can researchers access necessary material? Are readers better informed about source and reliability? Are disputes resolved more accurately and proportionately? These are proposed evaluation questions that make competing policy claims testable.

The difficult tradeoff is between preserving incentives and preserving the freedom to build on knowledge. Excessively weak protection can reward appropriation and undercut costly work. Excessively broad

protection can entrench incumbents and make ordinary learning legally hazardous. The objective should be a system in which creating, sharing, criticizing, adapting with permission, and independently improving all remain viable activities—not a system that maximizes one side’s control at any cost.

6.6 — A ninety-day programme for a small publisher or software team

An organization does not need to solve international copyright policy before improving its own practice. It needs a sequence of concrete decisions. The following programme is a proposed operating plan for a small team, not a compliance certification. Its timescale is illustrative and should be adapted to the complexity and risk of the work.

During the first month, establish the asset and use inventory. Mara’s team identifies manuscript versions, illustrations, quotations, publishing contracts, and AI-processing routes. Adrian’s team identifies contributors, dependencies, code-generation practices, documentation, datasets, and marks. Both select an accountable coordinator. They distinguish existing evidence from assumptions and stop obviously risky practices such as uploading confidential third-party material without a reviewed basis.

The first month’s output is not a perfectly clean archive. It is a prioritized map of gaps. Missing contractor assignments, uncertain image licences, unexplained code fragments, and unverified branding claims should become specific tasks. High-impact issues receive attention before low-impact administrative refinements. A team that discovers a problem should preserve evidence and obtain advice rather than quietly erase the history.

During the second month, build a workable release process. Establish which inputs may enter which AI services, under what configurations, and for which purposes. Add a rights review for new book editions and a component review for software releases. Create a record of required attributions and notices. Review the chosen marks in the intended

markets. Make the editorial or engineering contribution visible through sources, tests, corrections, and named responsibilities.

The process should be tested on actual examples. Give the reviewers a public-domain text in a newly protected translation, a short attributed but unauthorized extract, a clearly independent explanation, a copied code comment, and a truthful compatibility statement. Ask them to explain why the cases differ. A policy that only works when the answer is obvious will fail when teams most need guidance.

During the third month, test the dispute and correction routes. Simulate a complaint from an author, a licence concern from a customer, and a misleading official-product claim on a marketplace. Can the team retrieve the relevant version and permissions? Can it distinguish a factual allegation from a legal conclusion? Can it correct a genuine problem without making unsupported admissions about unrelated issues? A rehearsal exposes missing evidence more cheaply than a real dispute.

Measure practical indicators: the proportion of critical assets with reviewed provenance, unresolved issues at release, time needed to reconstruct a version's rights basis, accuracy of required notices, and time taken to correct verified errors. These are internal management measures, not proof of noninfringement. Avoid a target such as "zero similarity," which can reward pointless rewriting rather than lawful and useful production.

At the end of the programme, revisit the business model. Which customer value would survive a lawful independently produced substitute? Which value depends on verifiable quality, support, original investigation, or continuing maintenance? Which restrictions merely compensate for weak product design? These questions are commercially uncomfortable, but avoiding them can lead a company to overestimate what litigation will achieve.

The best achievable outcome is not invulnerability. It is a more defensible institution: aware of its rights, honest about uncertainty, careful with other people's work, capable of proving its own contributions, and prepared to compete where competition is lawful. That is a realistic

solution to many avoidable problems. It is not a promise to restore a world in which useful expression was expensive to reproduce.

Seminar — Four cases that resist easy answers

The purpose of this seminar is to practice separating questions before answering them. Each situation is fictional. The analyses are teaching approaches, not predictions of a particular court's decision. The relevant doctrinal sources are identified in the corresponding chapters. A good answer states the missing facts that could change the result; it does not simply repeat that "it depends."

Case A: the private book tutor. A university wants students to ask questions about a recent textbook. It owns printed copies. A contractor proposes scanning the book, placing it in a retrieval system, and allowing the assistant to answer from selected passages. The system is restricted to enrolled students. The university assumes that an educational purpose and the purchase of books settle the matter.

Begin with the acts, not the institution's good intentions. Scanning, storing, indexing, supplying passages to a model, and distributing answers require separate analysis. Identify the country, licence terms, relevant exceptions, authorized users, retention, and output behaviour. The fact that access is limited may matter without resolving every issue. The fact that the project is educational does not automatically import a US fair-use conclusion into another jurisdiction.

A safer proposed design might begin with licensed material, a clearly defined educational permission, or a reviewed exception applicable to the actual acts. It would limit access and retention consistently with that basis, preserve attribution, avoid systematic delivery of unauthorized replacement chapters, and test whether adversarial questions elicit long protected passages. These controls support a legal basis; they do not manufacture one where none exists.

Now alter the facts. The material is an openly licensed textbook, but some photographs are excluded from that licence. The answer changes at the component level. The institution can use the permissions actually

granted while handling the photographs separately. This is why “the textbook is open” should be verified against the licence and exceptions listed in the publication, rather than treated as a statement about every embedded asset.

Case B: the rival textbook with new words. A startup prompts a model to rewrite Mara’s entire book for younger readers. It changes every name, simplifies the language, and rearranges some sections. The resulting work retains the sequence of distinctive invented cases and their resolutions. It carries a different brand and credits Mara as an inspiration. The startup argues that the work is original because the wording is new.

The analysis should distinguish a general subject from expressive transformation. A derivative-work or adaptation question may remain despite low literal overlap. The credit does not supply permission. The new brand reduces one potential source-confusion concern but does not answer the copyright question. Review the particular expressive correspondences, access, permissions, and any claimed exception under the applicable law.

A genuinely independent educational treatment would start from the subject, primary evidence, and its own pedagogical choices, rather than from a command to preserve one book’s journey under different words. That process would provide stronger evidence of independence, but the actual output would still require review. A procedure is not a legal talisman. It should produce substantively different work, not just a cleaner story about how the work was made.

Now alter the facts again. The rival shares only conventional definitions and publicly established facts, with independent examples and structure. Mara’s sales fall. The remaining problem may be lawful competition rather than infringement. The appropriate response could involve improving discoverability, verified quality, or institutional distribution. A copyright complaint should not be used to hide that change in diagnosis.

Case C: the AI-written replacement library. Adrian wants to replace a third-party library whose licence no longer suits a planned product. An engineer supplies its complete source to a coding agent and asks for an equivalent implementation under the company's preferred licence. The resulting files contain new variable names and no copied licence header. All existing tests pass.

The tests establish a bounded behavioural result, not legal independence. The team must investigate whether protected expression was transformed, whether the initial processing was authorized, and what obligations follow from the material actually used. Removing a licence notice cannot be treated as a compliance step. Nor should an assumed need to change licences justify concealing the provenance of the replacement.

A more defensible route could use a legally available functional specification, qualified review of interoperability and other relevant rights, separately developed implementation choices, and an independent test design. The organization should examine whether the specification itself carries protected expression or confidential restrictions. It should also review the model's role without pretending to know its entire pre-training history. The goal is a documented, legally assessed implementation—not a claim of metaphysical freedom from all influence.

Now suppose the original library contains only an elementary implementation of a standard operation, and the new function follows from independent requirements. The assessment can be different. Open-source obligations do not arise from mere functional resemblance to every program under an open-source licence. This variation tests whether the reviewer can protect legitimate permissions and legitimate independence at the same time.

Case D: the authentic-looking fake edition. A marketplace seller publishes a machine-generated guide under a title and cover presentation suggesting that Mara approved it. The text is independently generated and contains no identifiable copied expression. The seller points to that absence and says the complaint must fail.

Copyright may not be the decisive instrument. Investigate the actual claims of authorship, endorsement, source, and affiliation, together with relevant trademark and other national protections. Preserve the listing and the context in which buyers encounter it. A response should target the misleading relationship rather than assert ownership over every independently written guide to Mara's subject.

Now replace the misleading presentation with an unmistakably independent critical guide. The author names Mara because her work is the subject of criticism. A claim based solely on that reference would require a different analysis. This variation tests whether a brand-protection process distinguishes deception from discussion. A creator should be able to defend a genuine identity without acquiring a veto over criticism.

Across all four cases, the same method survives: identify the asset, separate the acts, locate the applicable law, inspect permissions and exceptions, assess the actual correspondence, and choose a proportionate response. The method is deliberately more stable than the answer. New models and new cases will change facts and interpretations. They need not force every organization to restart from a slogan.

Epilogue — The freedom to learn from a book

Mara begins this story believing that a rival has taken something because the rival's product seems to do what hers does. Adrian begins with the same belief about software. By the end, neither has received a promise that the law will preserve every advantage created by hard work. They have something more useful: a sharper account of what can be protected, what must remain available to others, and where institutions need to do work that copyright cannot do alone.

That account can be uncomfortable for creators. A brilliant explanation can teach a reader enough to write another explanation. A useful program can reveal that a function is worth implementing. An effective method can spread beyond the book in which it was described. These are not merely defects in an otherwise perfect system of control. They are among the reasons people create, publish, and learn in the first place.

The discomfort also runs in the other direction. A system that transforms language can make appropriation harder to recognize. A model can conceal dependence behind fluent new sentences. A copied module can lose its licence notice while retaining protected structure. A fake edition can exploit a trusted name without copying a paragraph. Defending openness requires confronting these failures, not pretending that every transformation is independent creation.

The central distinction is therefore not between humans and machines. It is between justified and unjustified uses, truthful and misleading relationships, independent contributions and disguised dependence, effective competition and unlawful appropriation. Machines change scale, cost, and evidence. They do not make these distinctions unnecessary.

A mature response combines law, engineering, and institutional design. Law defines rights and limits. Engineering preserves versions,

permissions, and relevant evidence. Institutions support quality, attribution, maintenance, correction, and fair participation. None can substitute entirely for the others. A signature cannot decide originality; a judgment cannot ensure a product's reliability; a business model cannot excuse copying that the law prohibits.

There is no credible universal solution that makes every equivalent rewrite compensable, every output traceable to one source, or every creator economically secure. There are credible partial solutions to identifiable problems. The task is to combine them honestly, test their effects, and reject mechanisms whose apparent certainty depends on hiding the rights or interests of someone else.

The right to create should include meaningful protection against appropriation. The freedom to learn should include the possibility of understanding, criticizing, and doing something new with what one has learned. A culture that loses either side becomes poorer. The difficult work is to preserve both when imitation becomes cheap—and to remember that the purpose of a good book is not only to be bought, but also to make its reader capable of something the author does not entirely control.

Glossary — A working vocabulary

These definitions are study aids. Legal terms must be read in their statutory and jurisdictional context. Chapter discussions and the source register provide the relevant qualifications.

Adaptation or derivative work. A transformation of an existing work that may engage rights in its protected expression. Different language or format does not necessarily make the transformation independent.

Assignment. A transfer of specified rights. It differs from permission to use them and from ownership of a physical copy. Formalities and scope depend on the relevant law.

Attribution. Identification or acknowledgment of a source or creator. It can be required by law, licence, or professional norms, but does not itself supply permission for every use.

Chain of title. The documented basis connecting initial ownership to the party claiming the relevant rights, including employment rules, transfers, licences, and any intervening contributions.

Clean room. A process designed to separate examination of legally available reference material from implementation. It can improve evidence of independence but is not a universal legal safe harbour.

Commercial substitutability. The extent to which customers may choose one product instead of another. It is an economic relationship, not a synonym for copying protected expression.

Copyleft. A licensing approach attaching specified reciprocal conditions to covered uses, commonly including distribution of modified or combined covered software under defined terms.

Copyright. A legally delimited set of rights in qualifying expression. It is not general ownership of ideas, facts, methods, functions, or every useful result of reading a work.

Database right. In relevant systems, a right distinct from ordinary copyright that can protect qualifying investment in obtaining, verifying, or presenting database contents against specified acts.

Distinctiveness. A sign's capacity to identify a commercial source rather than merely describe or name the relevant category of goods or services.

Economic rights. Rights concerning specified exploitations of a work, such as reproduction, distribution, adaptation, or communication, as defined by the applicable law.

Exception or limitation. A legal rule allowing or excluding particular uses from an exclusive right under defined conditions. Its existence does not depend on a private licence for that use.

Expressive similarity. Correspondence in the protected form of a work, potentially beyond literal wording. Identifying it requires separating expression from unprotected or constrained elements.

Fair use. A contextual US doctrine evaluated through statutory factors. It is not a universal international permission or a fixed numerical allowance.

Functional equivalence. Agreement in relevant behaviour or practical capability. For software, the input domain and observable effects must be specified before the claim becomes testable.

General-purpose AI model. A regulatory category addressed by the EU AI Act. Its provider's regulatory obligations do not automatically settle the copyright status of every input or output.

Human authorship. The human creative contribution relevant to protection under applicable law. The use of an AI tool is not by itself disqualifying; accepting output is not by itself proof of authorship of all its expression.

Interoperability. The ability of systems or components to exchange information and use it meaningfully. Legal routes supporting interoperability have jurisdiction-specific limits.

Licence. Permission concerning specified rights or uses, under stated terms. A licensor cannot grant rights it does not hold, and a licence does not eliminate unrelated legal requirements.

Likelihood of confusion. A central trademark concept concerning whether relevant consumers may misunderstand commercial source or connection, assessed through the applicable legal framework and facts.

Moral rights. Author-related interests such as attribution and integrity. Their scope, transferability, waiver, and enforcement vary across legal systems.

Originality. The legal threshold concerning qualifying creative expression. It is not identical to novelty, scientific importance, effort, or commercial value.

Plagiarism. Misrepresentation of intellectual contribution or failure of required acknowledgment under relevant ethical or institutional norms. It can exist without copyright infringement, and vice versa.

Provenance. Information about origins, processing, and relationships among artifacts. A record can be useful evidence without conclusively establishing authorship, truth, or ownership.

Public domain. Material not subject to applicable exclusive copyright, including where protection does not arise or has expired. New components or versions may require separate analysis.

Retrieval-augmented generation. A system design that retrieves external material and provides it to a generative model while answering. It is not identical to changing model parameters through training.

Rights reservation. An expression retaining specified rights or excluding specified permissions, with effect determined by the relevant law and circumstances. Machine readability can matter in particular regimes.

Semantic equivalence. Agreement in conveyed meaning at a specified level. Meaning is not generally coextensive with the expression protected by copyright.

Software bill of materials. An inventory describing components in software. Standardized formats can support licensing and security work without proving the completeness or truth of every entry.

Text and data mining. Automated analytical processing of text or data. Legal exceptions addressing it differ in beneficiaries, purposes, access conditions, reservations, and permitted acts.

Trade secret. Information meeting the relevant legal requirements concerning secrecy, value, and protective measures. It is not a right over all useful information once publicly disclosed.

Trademark. A source-identifying sign protected under the relevant legal regime. Its function is different from ownership of the content, function, or idea associated with the product.

Source register

All web sources were consulted for the research cutoff of 11 September 2026. Links lead to the material used, not to a guarantee that the page or law will remain unchanged. Bracketed references in the text link here; each source title opens its linked text.

Legislation and judgments are distinguished from official guidance, policy reports, voluntary standards, and research proposals. WIPO Lex reproductions and national consolidations must be checked against later authoritative amendments when used for legal action. The book does not constitute a complete national-law audit.

[1] WIPO — Frequently asked questions: Copyright

Official explanatory guidance. Foundational scope, automatic protection, economic and moral rights.

[2] WIPO — Summary of the Berne Convention

Official treaty summary. National treatment, automatic protection, minimum rights, and terms.

[3] WTO — TRIPS Agreement, Part II, Sections 1 and 2

Treaty text. Articles 9–14 address copyright and related rights; Articles 15–21 address trademarks.

[4] WIPO — Summary of the WIPO Copyright Treaty

Official treaty summary. Digital-environment rights, computer programs, compilations, and protective measures.

[5] WIPO — Summary of the Paris Convention

Official treaty summary. National treatment, priority, industrial property, and unfair competition.

[6] WIPO — Madrid System

Official procedural overview of international trademark registration through designated members. Not a worldwide unitary trademark.

[7] WIPO — Nice Classification

Official classification resource for goods and services in trademark registration. Consult the edition applicable to a filing.

[8] United States Code — Title 17, Chapter 1

Statutory text, including sections 101, 102, 106, 107, and 117. Definitions, subject matter, exclusive rights, and limitations.

[9] United States Code — Title 17, Chapter 2

Statutory text, particularly sections 201–204. Ownership, physical copies, and transfers. Work-made-for-hire definitions also appear in section 101.

[10] United States Code — Title 17, Chapter 3

Statutory text. Duration, including special categories and historical rules.

[11] United States Code — Title 17, Chapter 4

Statutory text, particularly sections 408, 410, 411, and 412. Registration, litigation prerequisites, and remedial consequences.

[12] United States Code — Title 17, Chapter 5

Statutory text. Infringement, remedies, and section 512 service-provider provisions. Formal notices require jurisdiction-specific care.

[13] United States Code — Title 17, Chapter 12

Statutory text. Technological protection measures and copyright-management information; exceptions and exemptions require separate checking.

[14] US Copyright Office — Copyright and Artificial Intelligence, Part 2: Copyrightability

Official report, January 2025. Human authorship, expressive control, assistance, selection, arrangement, and modification. Agency analysis, not a new statute.

[15] US Copyright Office — Copyright and Artificial Intelligence, Part 3: Generative AI Training

Pre-publication report, May 2025. Training, acquisition, fair use, outputs, licensing, and market considerations. The pre-publication status is material.

[16] US Copyright Office — Copyright and Artificial Intelligence initiative

Official status and publication page consulted at the research cutoff. Part 3 was identified as pre-publication in the material retrieved.

[17] Andy Warhol Foundation for the Visual Arts, Inc. v. Goldsmith

US Supreme Court, 598 U.S. 508 (2023), decided 18 May 2023. Opinion concerning the first fair-use factor for the particular licensing use.

[18] Google LLC v. Oracle America, Inc.

US Supreme Court, 593 U.S. 1 (2021), decided 5 April 2021. Particular API-code reuse held fair use, assuming copyrightability without deciding it.

[19] SAS Institute Inc. v. World Programming Ltd

CJEU, C-406/10, 2 May 2012, ECLI:EU:C:2012:259. The linked material consulted is the Court's official press summary, No. 53/12, not the full judgment.

[20] Directive 2009/24/EC — Legal protection of computer programs

EU statutory text reproduced in WIPO Lex. Especially Articles 1, 5, 6, and 8. Subject matter, lawful observation, interoperability, and contractual limitations.

[21] Directive (EU) 2019/790 — Copyright in the Digital Single Market

EU statutory text reproduced in WIPO Lex. Especially Articles 3, 4, and 7. Text and data mining, reservations, and mandatory limits. National implementation also matters.

[22] European Commission — AI Act, Article 53

Official presentation of general-purpose AI model provider obligations. Copyright-compliance policy, training-content summary, and specified exemptions.

[23] European Commission — General-Purpose AI Code of Practice

Official explanatory resource. Voluntary compliance route published in July 2025; distinguish it from binding statutory duties.

[24] European Commission — AI regulatory framework

Official implementation and regulatory overview, consulted in September 2026. Application dates and transitional provisions should be rechecked before action.

[25] Regulation (EU) 2017/1001 — European Union trade mark

EU statutory text reproduced in WIPO Lex. Distinctiveness, infringement, limitations, classification, use, renewal, and territorial character.

[26] Directive 2006/116/EC — Term of protection

EU term directive, as amended, reproduced in WIPO Lex. General life-plus-seventy rule and category-specific provisions.

[27] Directive (EU) 2016/943 — Trade secrets

EU statutory text reproduced in WIPO Lex. Requirements for protection and distinctions between lawful and unlawful acquisition, use, and disclosure.

[28] Directive 2004/48/EC — Enforcement of intellectual property rights

EU statutory text reproduced in WIPO Lex. Enforcement measures, evidence, safeguards, and proportionality; national implementation remains relevant.

[29] Romania — Law No. 8/1996 on Copyright and Related Rights

Romanian consolidated text in WIPO Lex, amended through Law No. 69/2022 in the version consulted. Includes employee-software and mining provisions. Verify later amendments for transactions.

[30] Romania — Law No. 84/1998 on Trademarks and Geographical Indications

Romanian text in WIPO Lex, amended through Emergency Ordinance No. 168/2022 in the version consulted. Verify current consolidation and procedure before filing or enforcement.

[31] Swiss IPI — Copyright at a glance

Official national overview. Automatic protection and category-specific terms, including computer programs and photographs without individual character.

[32] Swiss IPI — Use of the trade mark

Official guidance on commercial use, five consecutive years of non-use, and cancellation procedures.

[33] Swiss IPI — Trade mark protection at a glance

Official overview. Registration, renewal, scope, and the need to investigate earlier rights independently.

[34] UK Government — Report on copyright and artificial intelligence

Government report, 18 March 2026. Policy direction, text and data mining, and computer-generated works. Proposals and discussion are not enacted amendments.

[35] Japan Agency for Cultural Affairs — General Understanding on AI and Copyright: overview

Official explanatory overview, May 2024. Article 30-4, non-enjoyment purposes, limits, and the distinction between development and use. Nonbinding interpretive guidance.

[36] USPTO — What is a trademark?

Official US introductory guidance. Source identification and distinctions among different identifiers and intellectual property rights.

[37] USPTO — Strong trademarks

Official US guidance on distinctiveness, including coined, arbitrary, suggestive, descriptive, and generic terms.

[38] USPTO — Likelihood of confusion

Official examination guidance. Similarity of marks and relatedness of goods and services; searches must extend beyond identical wording.

[39] USPTO — Trademark refusal: title of a single creative work

Official US registration guidance. Distinguishes single-work titles from qualifying series identifiers and discusses category-specific qualifications.

[40] Apache Software Foundation — Apache License, Version 2.0

Primary licence text. Permissions, redistribution conditions, notices, patent provisions, trademarks, and warranty limitations.

[41] Open Source Initiative — The MIT License

Primary licence text reproduced by OSI. Permissions, preservation of notices, and disclaimers.

[42] GNU General Public License, version 3

Primary licence text reproduced by OSI. Read the complete conditions and definitions rather than relying on the book's introductory explanation.

[43] GNU Affero General Public License, version 3

Primary licence text reproduced by OSI. Includes section 13 on remote network interaction for modified covered versions.

[44] Carlini et al. — Extracting Training Data from Large Language Models

Original research, arXiv:2012.07805; 2020 preprint, revised 2021. Demonstrates extraction from the studied GPT-2 setting, not universal recoverability from every model.

[45] IETF — RFC 9309: Robots Exclusion Protocol

Technical standard, September 2022. Crawler instructions are not access authorization or a security control.

[46] C2PA — Content Credentials Technical Specification, version 2.2

Primary technical specification. Signed provenance claims and trust model. A selected version, not a claim about the latest release or a legal ownership registry.

[47] W3C Community — Text and Data Mining Reservation Protocol

Community group materials and protocol reports. Machine-readable reservations and licensing information; not legislation or necessarily a W3C Recommendation.

[48] Bartz v. Anthropic PBC — Order granting final approval of settlement

US District Court, Northern District of California, case 24-cv-05417-AMO, document 680, 20 July 2026. Judicial order reproduced by the Association of American Publishers. A settlement, not a universal training precedent.

[49] Lee, Cooper, and Grimmelmann — Talkin’ ’Bout AI Generation

Copyright and the Generative-AI Supply Chain. Original legal scholarship, arXiv:2309.08133, first submitted 2023. Analytical framework, not binding law.

[50] Wang, Deng, Chiba-Okabe, Barak, and Su — An Economic Solution to Copyright Challenges of Generative AI

Original research proposal, arXiv:2404.13964, 2024. Contribution-based compensation using probabilistic and cooperative-game-theoretic methods.

[51] Creative Commons — Attribution 4.0 International

Primary legal code. Permissions, attribution and other conditions, reservations of other rights, and limits of the grant.

[52] SPDX — Overview

Official specification-project overview. Standardized component, licence, and related software information supports inventory and exchange, not automatic legal clearance.

[53] Dastar Corp. v. Twentieth Century Fox Film Corp.

US Supreme Court, 539 U.S. 23 (2003). Judicial opinion reproduced by Cornell LII. Limits on using trademark-origin claims as a substitute for copyright in communicative content.

[54] Feist Publications, Inc. v. Rural Telephone Service Co.

US Supreme Court, 499 U.S. 340 (1991). Judicial opinion reproduced by Cornell LII. Originality, facts, compilations, and the rejection of labour alone as the basis of copyright.

[55] Baker v. Selden

US Supreme Court, 101 U.S. 99. Judicial opinion reproduced by Cornell LII. Copyright in an explanation does not establish exclusive rights to use the system explained.

[56] Computer Associates International, Inc. v. Altai, Inc.

US Court of Appeals, Second Circuit, 982 F.2d 693 (1992). Judicial opinion reproduced by Justia. Software abstraction, filtration, and comparison.

[57] US Copyright Office — Fair Use Index

Official explanatory resource and index. Contextual four-factor analysis; no universal safe percentage or fixed allowance.

[58] WIPO — Frequently asked questions: Patents

Official introductory guidance. Exclusive rights, territoriality, licensing, and distinctions from copyright and secrecy.

[59] Directive 96/9/EC — Legal protection of databases

EU statutory text reproduced in WIPO Lex. Original framework, including Article 7 on qualifying investment. Read with later amendments, including Directive (EU) 2019/790, source 21.